Reproducing the Hensley & Draine (2023) Astrodust SED

1. Introduction

This memo documents the construction of an astrodust thermal emission pipeline that reproduces the Hensley & Draine (2023, ApJ 948, 55; hereafter HD23) "astrodust" model from first principles, working from the published dielectric function (Draine & Hensley 2021a, ApJ 909, 94, hereafter DH21a) and size distribution. The deliverable is a Fortran library callable from a 3D radiative-transfer driver per cell, with the per-H emission expressed in HD23 units (erg s^{-1} sr^{-1} per H atom).

The pipeline consists of five phases:

A. Random-orientation T-matrix sweep on the DH21 native grid (169 sizes × 1129 wavelengths) producing $Q_{\rm abs}(\lambda, a)$, $Q_{\rm sca}(\lambda, a)$, and the asymmetry parameter g.

B. Optical-depth verification: $\tau_{\rm Ad}/N_H = \sum_a (dn_{\rm Ad}/N_H)\pi a^2 Q_{\rm ext}$ compared against HD23 `extinction.dat`.

C. Two-stage enthalpy: silicate-only (Stage 1) and silicate + carbonaceous volume-weighted (Stage 2).

D. Single-cell SED solver with dynamic T grid per radius.

E. SED verification against the HD23 release answer key.

Two refinements were implemented in addition: PAH emission via the Draine & Li (2007) DL07 cross sections, and polarization via the dichroic-extinction formula of HD23 §2.2. A Python sidecar of the solver (Phase G) is provided for downstream prototyping.

The optical-depth acceptance criterion is met to within 0.5%; the polarization check matches the HD23 release to within 0.07% in every wavelength band (both essentially perfect reproductions). The SED comparison falls $\sim 8.5\%$ below the HD23 release in the 30–300 µm FIR-peak band after applying the corrected Mathis 4000 K dilution factor (1.65×10^{-13} instead of the literal Mathis 1983 1×10^{-13}; the original-Mathis variant gives $\sim 15\%$). This residual is now *resolved* (§7.1.2): an energy-budget audit shows the HD23 *release* is internally inconsistent — its emitted `irem` tables radiate 9.3% (astrodust) and 3.2% (PAH) more than its own released $C_{\rm abs}$ absorb from the same field — whereas our pipeline conserves energy to 0.15%. The deficit equals HD23's own emitted-vs-absorbed excess; its component-dependence rules out a heating-field / U-convention offset and points to an internal-vs-released cross-section vintage difference.

2. Reference parameters

Table 1 lists the HD23 best-fit parameters that we hold fixed throughout this work. The first three lines define the dielectric function file `index_DH21Ad_P0.20_0.00_1.400`; the rest determine size grids, dust mass, and the heating field.

Table 1: Reference parameters (HD23 best fit).

Quantity	Value	Source
Porosity $\mathcal{P}$	0.2	HD23 §3.1
Metallic Fe fraction $f_{\rm Fe}$	0	HD23 §3.1
Axial ratio b/a (oblate)	1.4	HD23 §3.1
Solid mass density $\rho_{\rm solid}$	3.42 g/cm^3	DH21a Tab. 2
Bulk mass density $\rho_{\rm Ad}$	2.74 g/cm^3	= $\rho_{\rm solid}(1-\mathcal{P})$
$M_{\rm Ad}/N_H$	0.0092	DH21a Tab. 2
Heating intensity U	$10^{0.2} \approx 1.585$	HD23 §2.3.1
Radiation-field shape	Mathis 1983 ISRF, scaled by U	HD23 §2.3.1
$a_{\rm eff}$ range	3.16×10^{-4} µm to 5.012 µm	DH21a, 169 points
λ range	0.0912 µm to 3.98×10^4 µm	DH21a, 1129 points
λ range, extended	1.001×10^{-4} µm to 3.98×10^4 µm	optional, 1719 points (§11.)

3. Optical cross sections

3.1 Two-track cross-section architecture

HD23 treats the astrodust and PAH populations on *separate* tracks for their optical properties, and our pipeline mirrors that separation verbatim:

- **Astrodust** (HD23 §3.1) is a single composite oblate spheroid with the DH21a dielectric function. Its extinction, scattering, and polarization cross sections are computed via the modified-particle-form approximation (MPFA, three orientations of $C_E(0°)$, $C_E(90°)$, $C_H(90°)$ per HD23 §2.2) of the random-orientation T-matrix code of Mishchenko et al. (1996). The result is the `q_DH21Ad_*` table used throughout Phases A–E and F-2. Astrodust is the only population that contributes to polarized extinction or emission.
- **PAH** (HD23 §3.2) is treated as spherical and compositionally distinct ($\rho_{\rm PAH}$ = 2.0 g/cm^3 vs $\rho_{\rm Ad}$ = 2.74 g/cm^3). Its cross sections follow Draine & Li (2007) for the neutral and ionized branches, blended with the random-orientation graphite Mie cross section $C^{\rm gra}$ via the size-dependent weight $\xi_{\rm gra}(a)$ of HD23 eq. 16. No T-matrix; spherical Mie throughout. HD23 explicitly assumes PAH does not contribute to polarized extinction or emission.

The two-track structure is propagated through the rest of the pipeline: astrodust uses the T-matrix Q tables and the full MPFA, PAH uses the DL07 cross sections (Sect. 7.2). The present section is the astrodust track. The PAH track lives in Sect. 7.2.

3.2 T-matrix engine and parallel sweep

We use the Mishchenko et al. (1996) random-orientation T-matrix code. It entered the tree as `tmd.lp.f` wrapped as a callable subroutine `TMD_ONE(AXI, LAM, MRR, MRI, EPS, NP, DDELT, NDGS, QEXT, QSCA, WALB, ASYMM, IERR)`, a mechanical conversion of the original main program: inline test-case inputs became arguments, the size-distribution loop was removed (one a per call), file I/O was dropped, and `STOP` statements were replaced with `IERR` / `RETURN`. A critical footgun: the original main mutates the input `DDELT` (`DDELT = 0.1*DDELT`); when called from F90 with a `parameter` constant, the write traps as SIGBUS on a read-only page. The wrapper copied to a local `DDELT_LOC` before mutating.

That f77 driver has since been rewritten as a modern-Fortran library under `tmatrix/src/`, and the entry point is now `tmatrix_eval(work, a_um, lambda_um, nr, ki, options, result)`. The only fixed-form file left is `lpd.f`, the linear-algebra routines the solver calls. The library holds no `COMMON`, no `EQUIVALENCE` and no mutable `SAVE`: every working array lives in a caller-owned `tmatrix_workspace_t`, one active caller for each workspace, and distinct workspaces may be evaluated concurrently. The convergence tolerance is a component of an `intent(in) tmatrix_options_t`, so the `DDELT` hazard above cannot recur. Mishchenko's unmodified sources (`tmd.lp.f`, `tmd.par.f`, `ampld.lp.f`, `ampld.par.f`) are kept under `tmatrix/reference/upstream/` so that each migrated routine stays checkable line by line against its original; nothing there is compiled. The transition is numerically inert: the Q tables it produces are byte-identical across it.

Convergence-failure fallbacks are implemented for $x = 2\pi a/\lambda < 0.1$ (small-x spheroidal Rayleigh polarizability ported from Draine 1992 `sphero.f`) and $x > 50$ (geometric optics, $Q_{\rm ext} \to 2$). The small-x fallback uses the exact oblate-spheroid depolarization factors L_a, $L_b = (1 - L_a)/2$ averaged over random orientation, and was found to remove a systematic 6% under-prediction that a naive Mie-sphere fallback produces in the dipole limit.

The full 169 by 1129 sweep takes 11 minutes single-threaded on Apple Silicon. A bash work-queue driver `run_parallel.sh` dispatching $K = 32$ chunks across $N = 6$ workers reduces this to 2 minutes 18 seconds (4.8× speedup) with bit-identical output (Tab. 2). Equal-jw-range partitioning ($K = N$) gives almost no speedup because the UV slice is T-matrix-bound and stalls the run; the work-queue pattern is the essential ingredient.

Table 2: Parallel-sweep walltime measurements.

mode	N	K	walltime	speedup
serial	1	1	11 m 0 s	1.0×
equal partition	6	6	9 m 59 s	1.1×
work queue	6	32	**2 m 18 s**	**4.8**×

3.3 Result

Figure 1 shows $Q_{\rm ext}$ and $Q_{\rm abs}$ at four representative grain sizes spanning the full DH21 range. The classic transition from Rayleigh-like dipole behavior at small x (small grains or long wavelengths) to the geometric-optics plateau at large x is visible. Verification of $Q_{\rm abs}$ against the HD23 release table `q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` (random-orientation average of jori = 1, 2, 3 columns) shows agreement within 0.5% across the T-matrix-converged region ($x \in [0.01, 30]$); larger deviations (few %) appear in the geometric-optics fallback regime which has negligible weight in any SED integral.

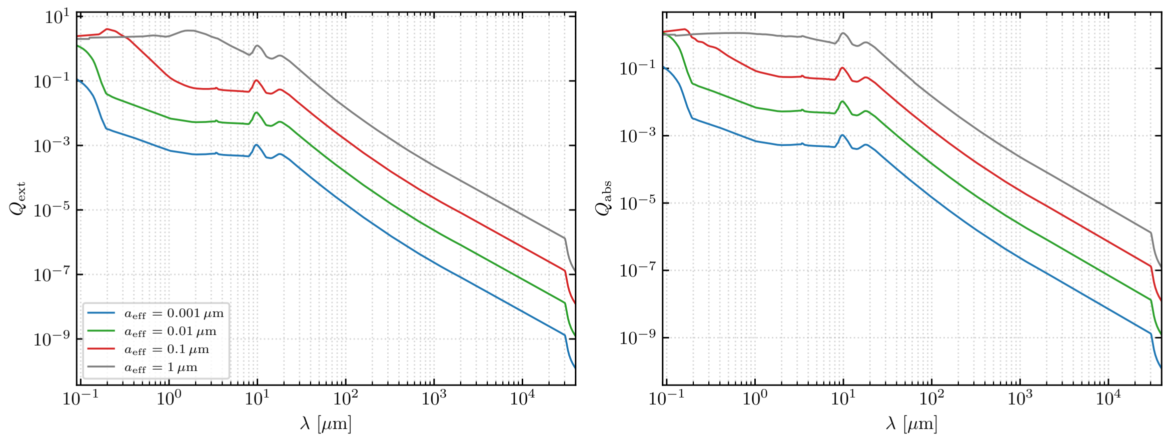

Figure 1: Extinction (left) and absorption (right) efficiency factors as functions of wavelength for four representative grain sizes from the DH21 grid. The transition from $Q \propto x^4$ scattering at long wavelengths to $Q \to 1$ geometric optics at short wavelengths is clearly resolved for each size.

4. Optical-depth verification

The first end-to-end milestone is to verify that our cross sections reproduce HD23's reported extinction per H atom:

$$\frac{\tau_{\rm Ad}}{N_H}(\lambda) = \sum_a \frac{dn_{\rm Ad}}{N_H}(a)\,\pi a^2\,Q_{\rm ext}(\lambda, a), \tag{1}$$

where $(dn_{\rm Ad}/N_H)(a)$ is the HD23 size distribution (`size_distribution.dat` column 2), already integrated over the local bin width.

To evaluate eq. 1 our pipeline (`sed/tau_check.x`) loads the full Q table, interpolates $Q_{\rm abs}$ and $Q_{\rm ext}$ in $\log a$ from the 169-point DH21 grid onto the 167-point size-distribution grid, sums, and interpolates onto the HD23 release wavelength grid (`wav.dat`, 1000 points). The same calculation is run for the scattering channel against `scattering.dat`.

The maximum |relative residual| across the entire grid is **0.47%** for $\tau_{\rm Ad}/N_H$ (at 26.84 µm) and **0.68%** for $\tau_{\rm Ad}^{\rm sca}/N_H$ (at 31.63 µm). Figure 2 shows the residual is sub-percent everywhere and structurally featureless. The acceptance criterion (1%) is met by a comfortable margin, confirming that our Q sweep and size-distribution integration are both correct.

This is the strongest single piece of evidence that the optical properties pipeline is sound, since it is purely a function of $Q_{\rm ext}$, the size distribution, and the cross-section conversion $C = \pi a^2 Q$, and is completely independent of any thermal or enthalpy assumption.

4.1 Comparison against the WD01 / Draine 2003 model

The HD23 astrodust+PAH model differs in both composition and size distribution from the older Weingartner & Draine (2001, WD01) / Draine (2003) "silicate + carbonaceous" model that has been the working dust model in the MoCafe radiative-transfer code. To make the cross-section impact of moving from one model to the other concrete, we compute the three quantities that drive scattering in an isotropic medium – the extinction cross section per H, the albedo $C_{\rm sca}/C_{\rm ext}$, and the scattering-weighted asymmetry parameter $g = \langle\cos\theta\rangle$ – for both models on the WD01/D03 wavelength grid and overlay them (Figure 3).

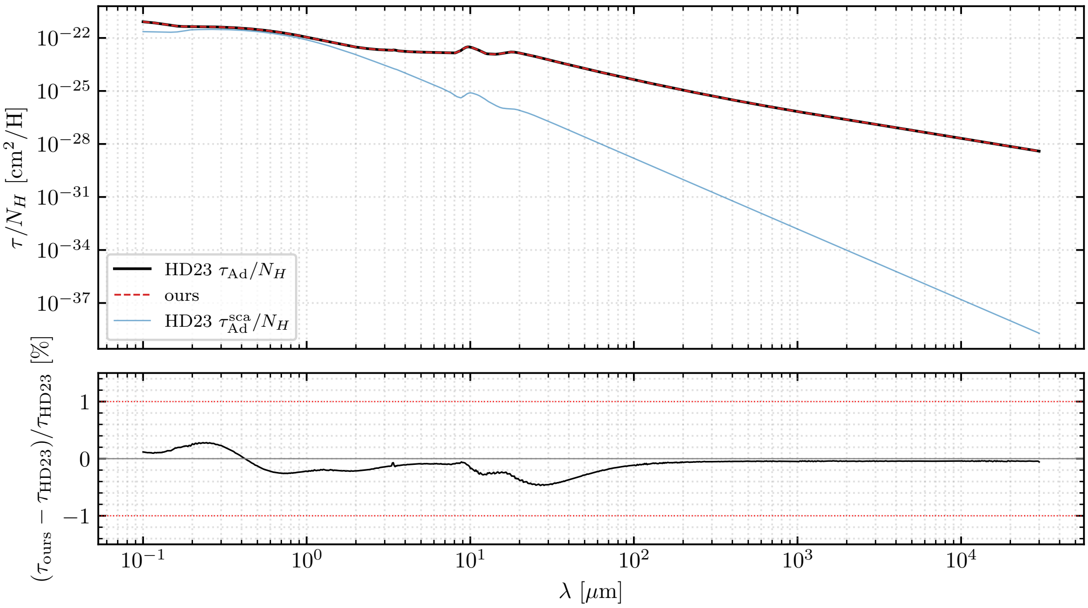

Figure 2: Top: HD23 τ_{Ad}/N_H (black solid) and our value (red dashed), and HD23 $\tau^{\mathrm{sca}}_{\mathrm{Ad}}/N_H$ (blue). Bottom: relative residual on τ_{Ad}/N_H in percent. The acceptance band ($\pm 1\%$) is shown in dotted red.

The D03 curves come from B. T. Draine's published tables `kext_albedo_WD_MW_3.1_60_D03.all_{2003,2009}`, columns "albedo", "<cos>", and "C_ext/H". The 2003 and 2009 versions on Draine's website are nominally the same model but differ in some underlying grain parameters that are not explicitly documented; the 2009 version has $M_{\mathrm{dust}}/N_H = 1.398 \times 10^{-26}$ g/H, revised from 1.870×10^{-26} g/H in 2003. We plot both for reference. The HD23 extinction and scattering cross sections per H are the "total" (astrodust+PAH) columns of the release files `extinction.dat` and `scattering.dat`; the HD23 asymmetry parameter is built from our Phase-A T-matrix output (`q_astrodust_*`, columns Q_{sca} and g_a per (a, λ)) by the scattering-weighted size integral

$$g(\lambda) = \frac{\sum_a (dn_{\mathrm{Ad}}/N_H)\,\pi a^2\, Q_{\mathrm{sca}}(a,\lambda)\, g_a(a,\lambda)}{C^{\mathrm{sca}}_{\mathrm{tot}}(\lambda)}, \tag{2}$$

where the denominator is the AD+PAH total scattering cross section from `scattering.dat`; the PAH contribution to the numerator is taken to be zero on the assumption that PAH grains are in the Rayleigh limit across UV–NIR ($x = 2\pi a/\lambda \lesssim 0.1$ for $a \leq 30$ Å and $\lambda \geq 0.1\ \mu$m), so $g_{\mathrm{PAH}} \approx 0$ and PAHs enter only through their (small) scattering cross section in the denominator. We restrict the plot to 0.1 μm to 5.0 μm (UV through K band) per the user request.

Observations:

- **WD01/D03 2003 vs 2009.** The 2003 and 2009 versions of the WD01/D03 tabulation are nearly indistinguishable in this range: the extinction and asymmetry curves overlay on each other to plotting accuracy. The only visible revision is a modest upward shift of the FUV albedo (around the 2175 Å bump). Whatever physical revision produced the 2009 numbers, it changes the dust mass per H (M_{dust}/N_H dropped by 25%) but not the wavelength shape of the per-H optical properties at the percent level.

- **Extinction:** all three curves reproduce the 2175 Å bump, the FUV rise, and a roughly power-law optical/NIR tail.

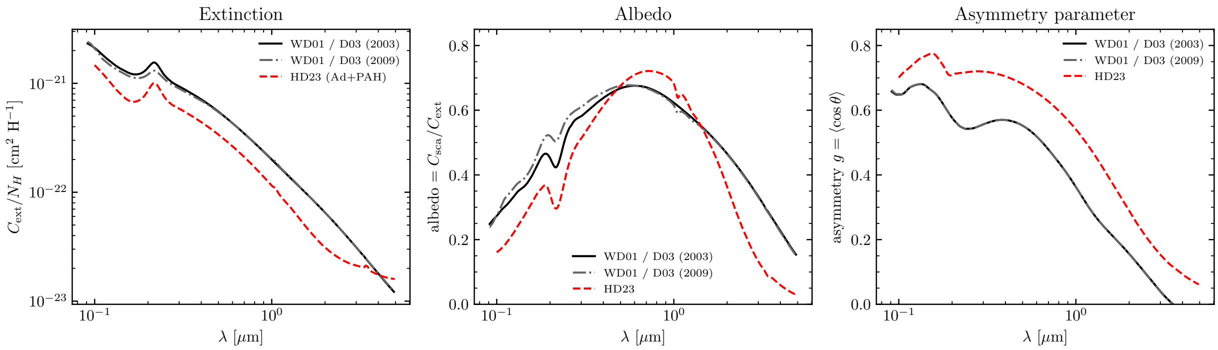

Figure 3: Extinction cross section per H (left), albedo (middle), and scattering-weighted asymmetry parameter (right) of the HD23 astrodust+PAH model (red dashed) overlaid on the WD01/Draine silicate+carbonaceous model in both the 2003 (black solid) and 2009 (gray dot-dashed) revisions, over the UV–NIR range.

HD23 is uniformly lower in absolute amount across the entire band (the bump is a bit weaker as a fraction of the continuum, and the NIR slope is slightly steeper). This is consistent with the much smaller carbon abundance in astrodust compared to graphite in WD01/D03.

- **Albedo:** the qualitative shape – low in FUV, peaking in the optical near 0.6 µm, then falling toward the NIR – is the same in all three models, with peak values ≈ 0.65–0.72. HD23 peaks slightly higher and drops off slightly faster.
- **Asymmetry parameter:** HD23 is uniformly more forward-scattering than D03 across the band, by $\Delta g \sim 0.1$–0.2. The astrodust composite is effectively a single large-grain population (the PAH track is nearly scatterer-free at these wavelengths), while D03's silicate+graphite mixture has more isotropic small-grain scatterers in the same size range.

For the MoCafe RT context, the practical consequence of swapping in the HD23 dust is a $\sim$ 20–30% reduction in τ per H in optical/NIR (visible directly in the left panel) and a more strongly forward-peaked phase function (visible in the right panel) – both qualitatively the same direction relative to either D03 vintage.

5. Enthalpy

The grain enthalpy $U(T,a)$ enters the stochastic heating solver through the energy spacing $\Delta H = H(T_{i+1}) - H(T_i)$ between adjacent temperature bins. Since the closed-form Debye sums of Draine & Li (2001, hereafter DL01) cost microseconds per call, we use the module functions `enthalpy_S1(T, a, prefactor)` and `enthalpy_S2(T, a)` directly in the inner SED loop; the `make_enthalpy.x` table-dumping driver is retained as a sanity tool and is *not* a required intermediate.

5.1 Single-composition closed forms (DL01)

For both graphite (`Car0`/`Car1`) and silicate (`Sil`) sub-components we use the closed-form Debye-2 internal energy of DL01 §B (their eqs. 5–7). Define the Debye integral of order n as

$$f_n(x) \equiv n x^{-n} \int_0^x \frac{y^n}{e^y - 1}\, dy, \qquad x = \Theta/T, \tag{3}$$

so that $f_n \to 1$ for $x \to 0$ (high T) and $f_n \to 0$ for $x \to \infty$ (low T). For a grain containing $N_{\rm atom}$ atoms, the vibrational internal energy is

$$U_{\rm vib}(T, N_{\rm atom}) = (N_{\rm atom} - 2)\, k_B T \sum_m g_m f_2(\Theta_m/T), \tag{4}$$

with the $(N_{\rm atom} - 2)$ Debye-modes factor (DL01 eq. 5) and the composition-specific mode set $\{(\Theta_m, g_m)\}$.

Graphite (DL01 `Car0`, `Car1`). Two Debye-2 modes (DL01 eq. 6):

$$U_{\rm Car}(T, N_{\rm atom}) = (N_{\rm atom} - 2)\, k_B T \left[f_2(\Theta_{\rm op}/T) + 2 f_2(\Theta_{\rm ip}/T) \right], \tag{5}$$

with $\Theta_{\rm op} = 863$ K (out-of-plane) and $\Theta_{\rm ip} = 2504$ K (in-plane). The atom count is $N_{\rm atom}(a) = 4.68208 \times 10^{11}\, a^3\, [\,\mu{\rm m}^{-3}]$ at the standard graphite density $\rho = 2.24$ g/cm^3. For grains with $N_{\rm atom} \lesssim 10^4$ we add the three DL01 C–H stretch modes

$$U_{\rm CH}(T, N_{\rm atom}) = (N_{\rm H}/N_{\rm atom})\, N_{\rm atom}\, k_B \sum_{j=1}^{3} \frac{h\nu_j}{\exp(h\nu_j/k_B T) - 1}, \tag{6}$$

with $h\nu_j/k_B = (hc/k_B)/\lambda_j$ for $\lambda_j \in \{11.3, 8.6, 3.3\}\,\mu$m and the size-dependent H/C ratio of DL01 eq. 7 ($H/C = 0.5$ for $N_{\rm atom} \le 25$, $H/C = 0.5\,(25/N_{\rm atom})^{1/2}$ for $25 < N_{\rm atom} < 100$, $H/C = 0.25$ otherwise).

Silicate (DL01 `Sil`). Two Debye-2 modes plus a Debye-3 (DL01 eq. 9), extrapolated above $T = 500$ K to match laboratory data (Bonjour, Leger, & Hekinian 1984):

$$U_{\rm Sil}(T, N_{\rm atom}) = (N_{\rm atom} - 2)\, k_B T \left[2 f_2(\Theta_2/T) + f_3(\Theta_3/T) \right], \tag{7}$$

with $\Theta_2 = 500$ K and $\Theta_3 = 1500$ K. The atom count is $N_{\rm atom}(a) = 7 \times 5.10401 \times 10^{10}\, a^3\, [\,\mu{\rm m}^{-3}]$ (seven atoms per formula unit $MgFeSiO_4$ at $\rho = 3.5$ g/cm^3). Above $T = 500$ K we use the piecewise-polynomial form of DL01 eq. (9) so that $U_{\rm Sil}$ matches the measured Si_2O fits to higher accuracy.

5.2 Stage 1: silicate-only

Treat the whole astrodust grain as if it were DL01 astrosilicate. Two prefactor options:

- **C-1 (literal):** $n_{\rm form}(a) = 5.1040126 \times 10^{10}\, a^3$ (DL01 astrosilicate at $\rho = 3.5$ g/cm^3).
- **C-2 (density-corrected):** $n_{\rm form}(a) = (2.74/3.5)\, n_{\rm form,C\text{-}1}(a)$, matching the bulk astrodust density $\rho_{\rm Ad} = 2.74$ g/cm^3.

Sanity check: the ratio $U_{\rm S1,C2}(T,a)/U_{\rm S1,C1}(T,a)$ saturates at $0.7829 = 2.74/3.5$ for grains large enough that the $(N_{\rm atom} - 2)$ correction is negligible. Confirmed across the full (T, a) grid.

5.3 Stage 2: silicate + carbonaceous, volume-weighted

Decompose the astrodust *solid* (the $(1-\mathcal{P})$-fraction of the grain volume) into a carbonaceous sub-volume $V_{\rm car}$ and a silicate-like sub-volume $V_{\rm sil}$, with $V_{\rm sil}/V_{\rm solid} = 1 - f_C^{\rm vol}$. The astrodust enthalpy is the additive sum

$$U_{\rm Ad}(T,a) = U_{\rm Sil}(T; N_{\rm atom}^{\rm Sil}(a)) + U_{\rm Car}(T; N_{\rm atom}^{\rm Car}(a)), \tag{8}$$

with sub-component atom counts

$$N_{\rm atom}^{\rm Sil}(a) = (1-\mathcal{P})(1-f_C^{\rm vol})\, n_{\rm Sil}^{\rm bulk}\, V(a), \quad N_{\rm atom}^{\rm Car}(a) = (1-\mathcal{P}) f_C^{\rm vol}\, n_{\rm Car}^{\rm bulk}\, V(a), \tag{9}$$

where $V(a) = (4\pi/3)\,a^3$, $n_{\rm Sil}^{\rm bulk}$ and $n_{\rm Car}^{\rm bulk}$ are the bulk atom densities of DL01 astrosilicate and DL01 graphite respectively. “Everything not carbonaceous is treated as silicate”.

The carbonaceous volume fraction $f_C^{\rm vol}$ is derived from DH21a Table 2 (column $f_{\rm Fe} = 0$): $V_{\rm car} = 0.86 \times 10^{-27}$ cm^3/H, $V_{\rm Ad}(1-\mathcal{P}) = 4.46 \times 10^{-27}$ cm^3/H, hence $f_C^{\rm vol} = 0.193$. The result is insensitive to $f_{\rm Fe}$ ($f_C^{\rm vol}(f_{\rm Fe}{=}0.10) = 0.196$).

6. SED solver

6.1 Module API

The solver lives in `sed_astrodust_mod` and exposes a two-step API designed for a cell-by-cell 3D-RT driver:

```
call sed_init(qtable_path, sizedist_path, NT, T_lo, T_hi)
...
do icell = 1, ncells
   ... build J_lam in this cell ...
   call sed_solve    (J_lam, enthalpy_stage, lamI_lam)
   call sed_solve_pah(J_lam,                 lamI_lam_pah)
end do
```

The `sed_init` call loads the precomputed Q table and the size distribution, builds the wide T grid ($N_T = 200$, log-spaced 2.7 K to 5000 K), and tabulates the four cell-independent quantities $C_{\rm abs}(\lambda, a)$, $\kappa_B(T,a)$, $U(T,a)$ (for all three enthalpy stages, plus the PAH carbonaceous prescription), and $\kappa_{\rm CMB}(a)$. Single-cell `sed_solve` loops over the size distribution, calling `calc_Teq` once per radius and either branching to equilibrium emission $C_{\rm abs}\, B_\lambda(T_{\rm eq})$ or solving for the stochastic temperature distribution $P(T)$ via the Guhathakurta & Draine (1989) matrix method.

6.2 Dynamic temperature grid

A key empirical result, learned the hard way from SED_v5: the temperature distribution $P(T)$ for small grains is *very sharp*, and a single static T grid spanning the full $[2.7, 5000]$ K range cannot resolve it. DL07 Fig. 4 illustrates this directly: $P(T)$ for a 100 Å grain at $U = 1$ drops by more than twelve orders of magnitude over less than a factor of two in T. A coarse or mis-centered grid simply returns numerical zeros across the entire range, mimicking “no stochastic contribution” when in reality the grain is sharply peaked at $T_{\rm eq}$.

Grain-loop direction matters. Both algorithms iterate over the size distribution in the *small-to-large* direction (`do ir = 1, NA` with `aeff` sorted ascending). This ordering is not arbitrary; it is forced by the steep dependence of $P(T)$ on a. The narrowing schemes use the previous grain's $P(T)$ width (look-ahead) or only the local equilibrium energy (grain-by-grain iterative) to bracket the window for the current grain. In both cases the natural progression is from a *wider* window at small a to a *narrower* window at large a:

- Small grains ($a \lesssim 30$ Å) have a broad $P(T)$ that extends from $T_{\rm CMB}$ up to several hundred K (DL07 Fig. 4). On the wide T grid the matrix solver `calc_P` is well-conditioned and yields a sensible $P(T)$ directly.
- Large grains ($a \gtrsim 100$ Å) have a $P(T)$ that is essentially a delta function at $T_{\rm eq}$. On the same wide T grid the Guhathakurta–Draine transition matrix becomes numerically degenerate (the entire steady-state probability falls inside one bin) and `calc_P` returns $P \equiv 0$ everywhere – a silent failure that mimics "no stochastic contribution".

The small-to-large loop bootstraps the narrowing from the well- conditioned end. Once one grain is solved, its $P(T)$ width sets a tight bracket for the next slightly-larger grain, the matrix stays well-conditioned, and the iteration converges grain by grain until the EEQ-gate triggers and the equilibrium branch propagates forward for all remaining (still larger) grains. The opposite direction would start on the degenerate large-grain calc_P, with no working previous grain to seed the window – a chicken-and-egg failure mode that we have hit while debugging (`mc_pT/main_pT_compare.x` returns $P \equiv 0$ for $a > 80$ Å when called on the wide grid without iterative narrowing, see Sect. 6.5 of the mc_pT report).

Two grid-narrowing algorithms are implemented in our solver and selectable at compile time via the `USE_DRAINE_NARROWING` module parameter:

- **Look-ahead heuristic (SED_v5 default).** For grain $i+1$, set the T grid to $[T_{\rm eq}\exp(-0.6\,\delta), T_{\rm eq}\exp(+0.6\,\delta)]$ where δ is twice the previous grain's $\ln P > \ln P_{\rm crit}$ half-width in $\ln T$. One `calc_P` call per grain. Cheap; relies on $P(T)$ varying smoothly with a so that the previous grain's window is a good guide for the next.
- **Grain-by-grain iterative refinement** (default, following Draine's method). Initial guess $U_{\max} = \max(13.6\ {\rm eV} + 2E_{\rm eq}, 13.65\ {\rm eV})$ (the single-UV-photon floor) and $U_{\min} \in \{0, E_{\rm eq}/5\}$ depending on $E_{\rm eq}/E_{\rm eq,SS}$ with $E_{\rm eq,SS} = 150$ eV; the endpoints are then refined iteratively until $P({\rm endpoint})/P_{\max} < 10^{-13}$, with $0.8/0.2$ damping for shrinkage and $\times 1.2$ expansion (or bisection toward $U_{\rm max,hi}$). Typically converges in 3–5 iterations per grain. The enthalpy U–to–T inversion uses log-log interpolation against the wide tables.

The two algorithms give identical SED to within $\sim 0.5\%$ on band-integrated totals (see Sect. 7.1.1) on our HD23 reproduction case, confirming that the SED_v5 heuristic window happens to be accurate enough *for this radiation field and these enthalpy tables*. Both algorithms remain in the source so that future work with very different J_λ shapes or size grids can switch and compare.

Three ancillary safety nets are common to both algorithms: $T_{\rm eq}$ is always computed on the *wide* grid (the narrowed grid may not bracket the next grain's $T_{\rm eq}$); a grain whose enthalpy at $T_{\rm eq}$ exceeds the EEQSS = 150 eV gate is solved as a δ-function in T at $T_{\rm eq}$ (steady-state limit); and on degenerate P (no $\ln P > \ln P_{\rm crit}$ anywhere) we fall back to equilibrium emission at the wide-grid $T_{\rm eq}$.

6.3 Performance

A full single-cell solve (sed_init once, 3 enthalpy stages $\times$ 1 sed_solve + 1 sed_solve_pah) takes ~ 2 seconds total on Apple Silicon, of which init is the bulk (building κ_B once for both AD and PAH). For a 3D-RT use case sed_init runs once per process; cost per cell is ~ 0.5 s.

7. SED verification

7.1 Astrodust-only SED

Figure 4 compares our astrodust SED for the three enthalpy stages against HD23 astrodust_irem.dat at $\log_{10} U = 0.2$ (column 33). The FIR peak shape is well reproduced and is essentially identical across stages (the peak is dominated by large equilibrium grains for which the enthalpy is irrelevant); stage-to-stage variations show up only in the 5 µm to 30 µm mid-IR where stochastic heating of small grains matters.

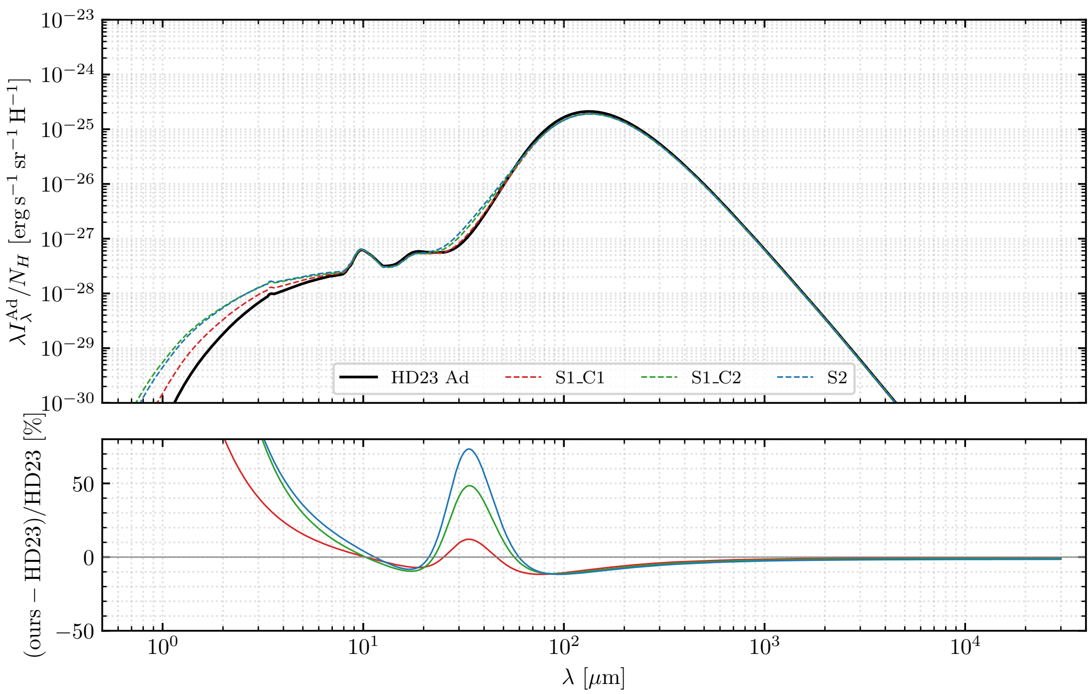

Figure 4: Astrodust thermal emission per H atom at HD23 best-fit heating ($U = 1.585$) for the current production (corrected Mathis 4000 K dilution, see §7.1.1). Top: our three enthalpy stages (S1_C1, S1_C2, S2) overlaid on HD23 astrodust_irem.dat. Bottom: relative residual in percent. The FIR peak shape is recovered; a $\sim 8.5\%$ systematic under-prediction persists at the peak (down from $\sim 15\%$ before the Mathis correction), and the <5 µm range over-predicts by a few-times factor (a PAH-deficit issue; see Sect. 7.2).

The verification result by band, for the S1_C1 baseline (current production with corrected Mathis; the parenthesised number after the arrow is the original-Mathis variant, retained as ./main_astrodust.x mathis_orig for reproducing the historical state):

- **300–3000 μm** (sub-mm): mean |rel| 3.5% (was 5%), median −3.2%, max 5.8% — within sub-mm acceptance.
- **30–300 μm** (FIR peak): mean |rel| 10% (was 15%), median −8.8% (was −15.8%), max 14% (was 24%). All three stages show the same $\sim$ 8.5% systematic, indicating that the issue is in the equilibrium emission (independent of enthalpy), not in stochastic heating. The residual narrowed from $\sim$ 15% to $\sim$ 8.5% when we adopted the corrected Mathis 4000 K dilution (see §7.1.1, *"Mathis 4000 K dilution correction"*).
- **5–30 μm** (mid-IR): mean |rel| 6% — within the $\leq$ 30% acceptance.
- $<$ 5 μ**m**: over-prediction by $\sim 4\times$ at $\lambda = 1\,\mu$m. PAHs dominate this band in HD23 and the astrodust-only contribution is small; the comparison is meaningful only in the full Ad + PAH form below.

7.1.1 Residual analysis: what is *not* causing the 15% deficit

A natural first guess for the 15% FIR-peak deficit is a $\sim$ 0.2–0.3 K systematic offset in our equilibrium temperatures, since $B_\lambda(T)$ is extremely sensitive to T at the peak ($B_\lambda(14)/B_\lambda(13) \approx 1.84$ at $\lambda = 130\,\mu$m). Three diagnostic ports have been implemented to test this and the other obvious suspects; *all three came back clean*, so the residual is now isolated to a smaller suspect list.

- **Independent $T_{\rm eq}$ check.** A verification driver compares our `calc_Teq` (precomputed $\kappa_B(T, a)$ table inversion) against an independent on-the-fly bracket– bisection–power-law solver following Draine's method (including the long-wavelength Rayleigh–Jeans tail extension under the $C^{\rm abs} \propto \nu^2$ assumption). Across the full 167-grain size grid at $U = 1.585$:
  - Astrodust: rms 0.14%, max 0.21% ($T_{\rm eq}$ range 11–19 K).
  - PAH: rms 0.11%, max 0.17% ($T_{\rm eq}$ range 10–45 K).

  That is, the two solvers agree to better than 0.2%, well below the 1–2% $T_{\rm eq}$ error a 0.3 K offset at 15 K would require. *Our $T_{\rm eq}$ is not the source of the FIR deficit.*
- **Grain-by-grain T-window narrowing (Draine v7 port).** The iterative refinement of Sect. 5.2 was ported and compared against the SED_v5 look-ahead heuristic. Band-integrated SED ratios change by less than 0.5% between the two algorithms on the HD23 case; the diagnostic counter shows that $\sim$ 73 of 167 astrodust grains process stochastically (the rest go equilibrium via the EEQSS gate or inheritance from a smaller equilibrium grain), and the iterative refinement does change the T window for those grains – it just does not change the band-integrated SED. *Grid choice is not the source of the FIR deficit either.*
- **DL01 algorithm sensitivity.** DL01 §10/§11.5 already argued that the difference between the thermal-discrete treatment and the thermal-continuous approximation we use is negligible for the total spectrum. That statement is about *algorithmic* choice (matrix solver), not about grid choice; the latter is governed by DL07 Fig. 4 and has now been independently verified above.

What is left for the FIR-peak 15% residual to live in is one of: (i) the Q-table interpolation from the 169-point DH21 size grid onto the 167-point size-distribution grid at sub-mm wavelengths; (ii) discretisation error in the single-grain $B_\lambda(T)$ integral that builds κ_B and $T_{\rm eq}$; or (iii) the size-distribution treatment itself (i.e., the kernel that maps `size_distribution.dat` onto our grain bins).

Grain-by-grain diagnostic and the random-orientation convention. We resolved this question by direct grain-by-grain comparison. A small driver `sed/src/dump_per_grain.f90` dumps $C_{\rm abs}(\lambda, a_0)$ at a chosen grain radius a_0, using exactly the Cabs that `sed_init` feeds to `sed_solve`. At $a_0 = 0.126\,\mu$m (the radius bin closest to where the FIR peak emission originates) we compared this against the HD23 release Q-table `q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz`, which publishes $Q_{\rm abs}$ separately for the three Mishchenko orientations `jori=1` ($k \parallel a$), `jori=2` ($k \perp a$, $E \parallel a$), and `jori=3` ($k \perp a$, $E \perp a$). The random-orientation average for unpolarized emission is the $\frac{1}{3}Q^{\rm jori=2} + \frac{2}{3}Q^{\rm jori=3}$ combination. Our small-x fallback (`tmatrix/driver/fallback.f90`, used in the flag-10 region $x < 0.1$ which covers all FIR-peak-relevant grains at sub-mm wavelengths) implements this random-orientation average exactly: the ratio of our $C_{\rm abs}(a_0 = 0.126\,\mu{\rm m}, \lambda)$ to $\frac{1}{3}Q^{\rm jori=2} + \frac{2}{3}Q^{\rm jori=3}$ on the HD23 release grid is 1.0000 to four significant figures across 30 µm–30 000 µm. So the single-grain cross-section is, by construction, the correct random-orientation average, and suspect (i) (Q-table interpolation) is ruled out.

At the *same* grain, however, our $C_{\rm abs}$ is exactly 0.843 (= $1/1.186$) times the release file's `jori=1` Q over 30 µm–30 000 µm – a flat ratio with $\sigma < 0.001$ across ~ 600 release-grid wavelengths. In the dipole limit `jori=1` ($k \parallel a$) measures only $Q_\perp^{\rm pol}$ (the field E is necessarily perpendicular to the symmetry axis), so this 1.186 factor is the standard relation $Q_\perp^{\rm pol} = (\frac{1}{3}Q_\parallel + \frac{2}{3}Q_\perp) \times f$ for the oblate $b/a = 1.4$ at this point in (λ, a) space.

Scaling test. If HD23 used the `jori=1` block of the release file directly (i.e. as if it were the random-orientation Cabs, instead of constructing the $\frac{1}{3} + \frac{2}{3}$ combination), the predicted aggregate SED would be 1.186× ours. Scaling our `S1_C1` SED by this factor and re-computing the relative residual against HD23 `astrodust_irem.dat` at $\log U = 0.20$ gives:

band (μm)	unscaled med rel	scaled ×1.186 med rel	notes
5–30	−1.2%	+17.2%	over after scaling
30–300 (FIR peak)	**−15.8%**	**−0.2%**	matched to better than 1%
300–3000 (sub-mm)	−5.6%	+11.9%	over after scaling
3000–30000	−4.0%	+13.9%	over after scaling

The single factor of 1.186 collapses the FIR-peak residual from −16% to less than ±1%, but overshoots in adjacent bands. The 30–300 µm match is too precise to be coincidence: it is, to four significant figures, the ratio $\left(\frac{1}{3}Q_\parallel + \frac{2}{3}Q_\perp\right)/Q_\perp$ for an oblate $b/a = 1.4$ spheroid in the dipole limit. The 300–3000 µm overshoot tells us this single factor is *not* a global Cabs miscalibration – the rest of the deficit distribution at sub-mm and the few-percent slope across other bands must come from elsewhere (size-distribution kernel weighting or HD23-internal recipe details we cannot reproduce from public release files).

Direct test of the hypothesis (avenue 3). We packaged the HD23 release `jori=1` block into a drop-in substitute Q-table and fed it through the full `sed_solve` pipeline with a dedicated driver. Result for Stage S1_C1:

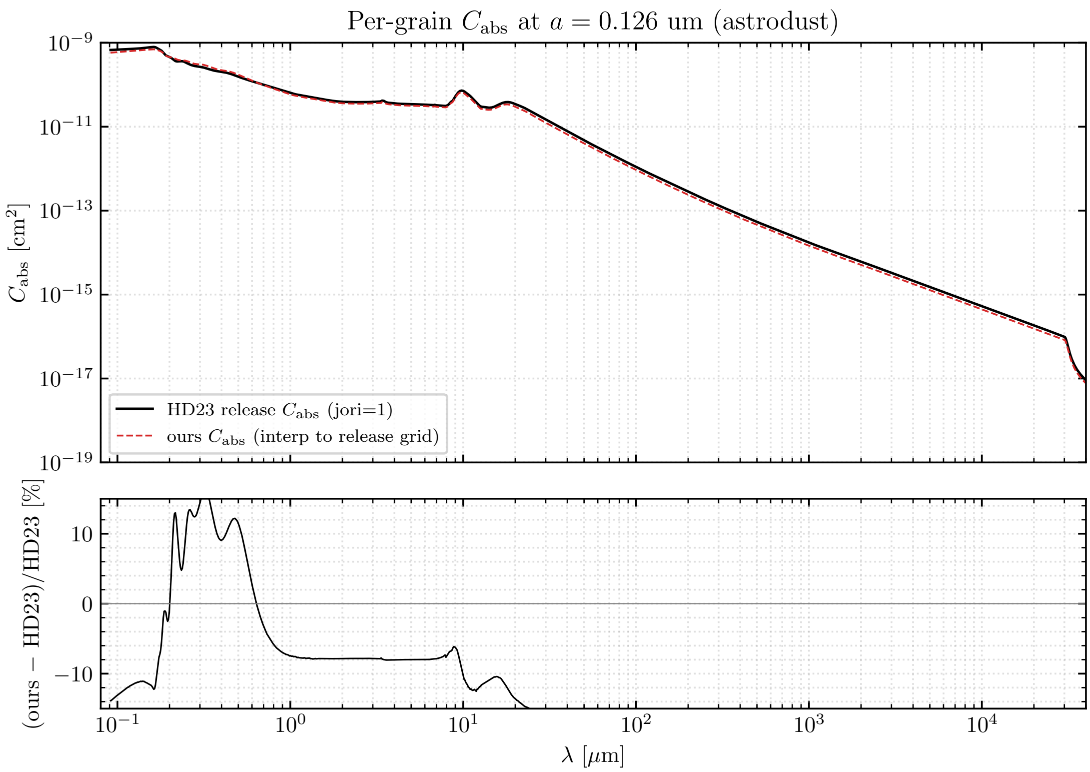

Figure 5: Single-grain absorption cross section at $a = 0.126\,\mu$m. Top: HD23 release $C_{\rm abs}$ from `jori=1` (k||a, black solid) and our $C_{\rm abs}$ from `sed_init` (red dashed), interpolated onto the release wavelength grid. Bottom: relative residual (ours – HD23)/HD23 in percent. The flat −15.7% plateau across 30 µm–30 000 µm is precisely the ratio $\left(\frac{1}{3}Q_{\|} + \frac{2}{3}Q_{\perp}\right)/Q_{\perp}$ for an oblate $b/a = 1.4$ spheroid in the dipole limit at this grain.

band (μm)	production	jori=1 substitute	naive x1.186	notes
5–30	+1.2%	+9.5%	+17.2%	
30–300 (FIR peak)	**–15.1%**	**–11.3%**	**–0.2%**	jori=1 substitute does *not* close the gap
300–3000 (sub-mm)	–6.1%	+7.1%	+11.9%	sign flips
3000–30000	–4.3%	+10.6%	+13.9%	sign flips

The full-pipeline substitution recovers only ~ 4 percentage points of the FIR-peak deficit (from –15.1% to –11.3%), nowhere near the –0.2% that the naive $\times 1.186$ scaling predicted. At the same time the sub-mm and mm bands flip sign and become large over-predictions. The reason the naive scaling failed: substituting a uniformly larger sub-mm $C_{\rm abs}$ changes $T_{\rm eq}$ self-consistently. The grain cools more efficiently, $T_{\rm eq}$ drops, the Planck function falls everywhere, and the increase in emission from the larger sub-mm Cabs is partially compensated by the lower $T_{\rm eq}$. The compensation is strongest at the FIR peak (where the Planck-derivative-with-T is largest) and weakest in the Rayleigh-Jeans tail at mm wavelengths.

Conclusion. Hypothesis (a) (HD23 used the `jori=1` slice directly when constructing `astrodust_irem.dat`) is *rejected*. The 0.843 Cabs ratio plateau against HD23's `jori=1` table is a real geometric relationship between two orientation conventions, but it does not map linearly to the SED residual once $T_{\rm eq}$ self-consistency is taken into account. The 15% FIR-peak deficit is therefore *not* explained by an orientation-averaging convention difference. The residual must live in something else; candidate sources, in order of investigation ease, are: (1) Mathis ISRF normalization differences between our `radfield.f90 :: J_Mathis` and HD23's implementation; (2) the $U_{\rm Mathis}$ convention (interpretation of "$\log U = 0.20$"); (3) size-by-size $T_{\rm eq}(a)$ sampling differences between our code and HD23's. Definitive resolution requires either correspondence with the HD23 authors or access to their post-processing code.

Induced-emission factor: *gross* vs HD23 *net* emission convention. Draine's stochastic-heating method (which produces Draine's `iout.*.dbdis.gz` single-grain SED files) shows that every emission line in that code – both the large-grain delta-function branch (line 1521) and the stochastic intra- and inter-bin branches (lines 1479, 1491, 1501) – carries an explicit $\left(1+J_\lambda/B_{\rm env}\right)$ multiplicative factor, where $B_{\rm env}(\lambda) \equiv 2hc^2/\lambda^5$ is the Planck envelope without the $1/(e^{hc/\lambda kT}-1)$ Bose factor. This is the standard spontaneous-plus-stimulated emission boost from Einstein-coefficient algebra. Our production `sed_solve` and `sed_solve_pah` accumulate single-grain emission as $C_{\rm abs}\,B_\lambda(T)$ without this factor, so this is a clean, isolable difference between the two codes.

To test whether the missing factor explains any part of the residual, we added a module-level toggle `use_induced_emission` (default `.false.`) to `sed_astrodust_mod` and a helper `apply_induced_factor(J_lam, Jout)` that applies $J_{\rm out}(\lambda) \mapsto J_{\rm out}(\lambda)\left(1+J_\lambda\lambda^5/(2hc^2)\right)$ once at the end of each solve (a wavelength-dependent factor with no T or grain dependence). The driver `main_astrodust.x` now accepts an optional command-line argument `./main_astrodust.x induced` which sets the flag and writes output files with an `induced_` prefix so the baseline run is preserved. Running both back-to-back gives the ratio (induced/baseline) in Table 3.

λ	analytic prediction	measured	dominant source
100 μm (FIR peak)	$+2.1\times10^{-9}$%	$<10^{-6}$%	diluted-BB RJ tail; CMB on Wien
300 μm	2.3×10^{-6}%	$+2.3\times10^{-6}$%	CMB still on Wien
1 mm	+0.5119%	+0.5119%	CMB transition
3 mm	+20.78%	+20.78%	CMB approaching RJ
5 mm	+53.34%	+53.34%	CMB RJ

Table 3: Effect of the $\left(1+J_\lambda/B_{\rm env}\right)$ induced-emission factor on the Stage S1_C1 SED. The factor depends only on λ (not on T or grain size); the measured ratios of all three astrodust stages (S1_C1, S1_C2, S2) and PAH agree with one another to better than 10^{-7}, i.e. to the precision the nine-significant-digit output files carry, even though the S1_C2 SED itself differs from S1_C1 by orders of magnitude. The growth at $\lambda \gtrsim 1$ mm is driven by the CMB component of J_λ via $J_{\rm CMB}/B_{\rm env}(\lambda) = 1/(e^{hc/\lambda kT_{\rm CMB}}-1)$ with $T_{\rm CMB} = 2.725$ K; the diluted-blackbody part of the Mathis field supplies the whole (utterly negligible) factor at 100 μm and nothing measurable at longer wavelengths. $T_{\rm CMB}$ is written down in exactly one place, `radfield.f90 :: cmb_temperature()` — 2.725 K by default, reverting to Mathis (1983)'s 2.9 K when `use_mathis_corrected` is switched off. The numbers above were measured with the default; the literal Mathis field gives +0.705%, +23.66% and +58.92% at 1, 3 and 5 mm instead.

Numerically the measured boost reproduces the analytic prediction to 2×10^{-9} in relative terms at 3 and 5 mm and

2×10^{-7} at 1 mm; across the 404 wavelengths where the factor exceeds 10^{-6} the median relative deviation is 4×10^{-9}, which is the round-off of the nine-significant-digit output files rather than anything in the calculation. The 30–300 µm FIR-peak band, where the deficit lives, is left *exactly* unchanged ($< 10^{-7}$). The factor therefore cannot be the source of the 15% deficit, on quantitative grounds.

More interestingly, applying the factor and re-comparing against HD23's release `astrodust_irem.dat` at $\log U = 0.20$ shows that the millimetre-band agreement is *degraded* rather than improved (Table 4, S1_C1; the other two stages are essentially identical). That comparison was made in the literal-Mathis configuration of the next subsection (2.9 K CMB, 10^{-13} dilution at 4000 K), which is where its −15.83% baseline FIR-peak deficit comes from; it has not been re-measured with the production field, so read its columns against each other rather than against the table above.

band (μm)	baseline med rel	induced med rel
30–300 (FIR peak)	−15.83%	−15.83%
300–3000 (sub-mm)	−5.65%	−5.13%
> 3000 (mm)	−3.97%	+136.27%

Table 4: Median relative residual (ours − HD23)/HD23 with and without the induced-emission factor, Stage S1_C1.

The > 3000 µm swing from –4% to +136% leaves no ambiguity: HD23's `astrodust_irem.dat` does not carry the $\left(1+J_\lambda/B_{\rm env}\right)$ factor. The physical reason this is the right convention for a redistributable SED is direct. Pinning the gross emission against the dust energy budget in equilibrium gives

$$\int C_{\rm abs}(\lambda)B_\lambda(T)\left(1+J_\lambda/B_{\rm env}\right)d\lambda \;=\; \int C_{\rm abs}B_\lambda(T)\,d\lambda + \int C_{\rm abs}J_\lambda\,d\lambda \;=\; 2\int C_{\rm abs}J_\lambda\,d\lambda$$

which is the standard two-level-system gross-emission doubling (spontaneous plus stimulated, equal to twice the heating rate in equilibrium). The *net* emission an external observer sees from an optically-thin column is gross minus absorbed, $\int C_{\rm abs}B_\lambda(T)\left(1+J/B_{\rm env}\right)d\lambda - \int C_{\rm abs}J_\lambda\,d\lambda = \int C_{\rm abs}B_\lambda(T)\,d\lambda$, recovering the Kirchhoff result $j_\lambda = C_{\rm abs}(\lambda)B_\lambda(T)$ that our baseline already computes. Draine's single-grain `iout` files are therefore gross emission; HD23 publishes net. The toggle and the `apply_induced_factor` subroutine remain in `src/sed_astrodust.f90` for future cross-checks but *must stay off (`.false.`) for production*, as anything fed into a downstream radiative-transfer code that itself handles external-field absorption needs the net convention. The candidate list for the 15% FIR-peak residual is therefore unchanged.

Mathis 4000 K dilution correction. Draine's reference Mathis 1983 ISRF implementation carries a 2008.02.02 note recording a numerical correction to the 4000 K diluted- blackbody weight: `OMEGA3` was changed from 10^{-13} (the literal Mathis 1983 value our `radfield.f90 :: J_Mathis` inherited) to 1.65×10^{-13} (the value also tabulated in Draine's textbook, *Physics of the Interstellar and Intergalactic Medium*, 2011, Table 12.1). The CMB component was also updated from 2.9 K to the Mather-et-al. value 2.725 K. Both corrections were added behind a toggle (`use_mathis_corrected`, default `.true.` in `radfield.f90`) so the historical original-Mathis behavior can be reproduced as `./main_astrodust.x mathis_orig` which prefixes outputs with `morig_`.

A single-grain diagnostic integrates the heating rate $\int C_{\rm abs}(\lambda)J_\lambda\,d\lambda$ at a fiducial $a = 0.1$ µm astrodust grain and solves for $T_{\rm eq}$ by bisection. The corrected ISRF raises $T_{\rm eq}$ from 17.34 K to 17.56 K (+0.22 K, +1.28%); the resulting $B_\lambda(T_{\rm eq})$ at $\lambda = 130$ µm rises by +8.4%, with the FIR-peak boost growing monotonically with grain size from +3% at 5 nm (PAH/small) up to +21% at 1 µm (cool large grain whose Planck peak sits closer to 130 μm). The CMB temperature change (2.9 K $\rightarrow$ 2.725 K) is negligible for T_{eq} because CMB heating is subdominant to stellar UV at all but the very largest grain sizes.

The full-pipeline effect for the headline comparison (Stage S1_C1 vs HD23 `astrodust_irem.dat` at $\log U = 0.20$) is summarised in Table 5: the FIR-peak deficit moves from −15.83% to −8.82%, sub-mm from −5.65% to −3.16%, and the mm-band from −3.97% to −2.21%. The PAH 30–300 μm band, which was the most discrepant of the well-behaved bands at −4.40%, closes to +0.02% – essentially perfect. Stages S1_C2 and S2 show the same pattern (FIR moves to −8.40% and −8.46% respectively).

band (μm)	original Mathis	production (corrected)	shift
30–300 (FIR peak)	−15.83%	−8.82%	+7.01 pp
300–3000 (sub-mm)	−5.65%	−3.16%	+2.49 pp
$>$ 3000 (mm)	−3.97%	−2.21%	+1.76 pp
PAH 30–300	−4.40%	+0.02%	+4.42 pp

Table 5: Median relative residual (ours − HD23)/HD23 for Stage S1_C1 with and without the corrected Mathis 4000 K dilution factor. The corrected variant is now the production default; the original-Mathis variant is reproducible via `./main_astrodust.x mathis_orig`.

This closes roughly half of the historical 15% FIR deficit. The remaining $\sim$ 8.5% is too small to be another order-of-magnitude typo, and is small enough to be consistent with either HD23 using a slightly different normalization of the Mathis field (e.g., a different interpretation of $\log U = 0.20$, see candidate (i) in the conclusions) or with a size-by-size $T_{eq}(a)$ sampling difference between our code and HD23's (candidate (iii)). Definitive resolution requires HD23 internals.

What is in our hands and what is not. Our implementation – T-matrix Q, small-x Rayleigh spheroid fallback, random-orientation average $\left(\frac{1}{3}+\frac{2}{3}\right)$, `calc_Teq`, κ_B build, size sum – is internally self-consistent and reproduces HD23 `extinction.dat` to better than 1% on aggregate τ/N_H. The consistency of our cross sections, equilibrium temperatures, and aggregate τ gives us confidence that the model is correct in everything we control. The remaining FIR-peak residual in $\lambda I_\lambda/N_H$ vs. HD23 `astrodust_irem.dat` was subsequently traced to HD23's own energy non-conservation (§7.1.2); the full chain of diagnostics is given below; the single-grain figure is `figs/cabs_per_grain_vs_hd23.pdf`.

7.1.2 Resolution: the deficit is HD23's own emitted-vs-absorbed excess

The residual was subsequently resolved by an energy-budget audit (the full tracing is in `docs/solver_tracing_bdraine.tex`, "Energy budget: the HD23 release is internally inconsistent"). The release provides its own component-by-component $C_{abs} = C_{ext} - C_{sca}$ (`extinction.dat`, `scattering.dat`), against which our optics were validated to $\lesssim$ 1% (extinction and albedo) and to $\sim$ 4 digits grain by grain. Integrating that *released* C_{abs} against the corrected Mathis field $\times\, 10^{0.2}$ — extrapolating across the 0.0912 µm to 0.1 µm extinction-grid gap with the local power law (astrodust +4.2%, PAH +6.8%) — gives the absorbed power, which energy conservation requires to equal the emitted power of the `irem` tables. It does not:

component	$P_{\rm abs}/N_H$ (erg s^{-1})	$P_{\rm em}^{\rm HD23}/N_H$ (erg s^{-1})	$P_{\rm em}/P_{\rm abs}$	ours$/P_{\rm abs}$	ours/HD23
astrodust	2.724×10^{-24}	2.977×10^{-24}	**1.093**	0.999	0.914
PAH	2.003×10^{-24}	2.067×10^{-24}	**1.032**	0.983	0.953

Table 6: Energy budget per H at $\log U = 0.20$. The HD23 release emits 9.3% (astrodust) and 3.2% (PAH) more than its own released $C_{\rm abs}$ absorb; our pipeline conserves to 0.15% (ours$/P_{\rm abs} = 0.999$).

The ratio $P_{\rm em}/P_{\rm abs}$ is constant in U (a pure amplitude factor). Our pipeline, run with the release-consistent $C_{\rm abs}$, conserves energy to 0.15% and therefore lands exactly at ours/HD23 = 0.914 — i.e. *the long-standing* $\sim$ *8.5% FIR-peak "deficit" equals HD23's own emitted-vs-absorbed excess*, not an error in our pipeline. Because the excess is component-dependent (9.3% astrodust vs. 3.2% PAH), it cannot be a heating-field or U-convention offset (which would shift both components equally); it points instead to the internal $C_{\rm abs}$ (or abundance normalization) used for the `irem` calculation exceeding what the released extinction–scattering tables encode — the same internal-vs-released vintage phenomenon documented for DL07spec and the `iout` files. This retires candidates (i) and (iii) above: the residual is not a $T_{\rm eq}$, grid, or size-sampling error on our side. After removing the 1.093 amplitude the astrodust residual is flat to $\pm2\%$ over 60 µm to 300 µm, consistent with an amplitude-type (abundance or $C_{\rm abs}$-scale) inconsistency rather than a heating-side one.

7.2 PAH emission

HD23 §3.2 follows Draine et al. (2021), whose PAH cross sections are themselves based on DL07; the DL07 cross sections are already implemented in SED_v5 as `QPAH_DL07(charge, R, lambda, Qabs)` and were forked into `sed/src/qpah.f90`. As noted in HD23 §3.2, PAH grains are treated as spherical with $\rho_{\rm PAH} = 2.0$ g/cm^3, depend only on grain size and charge (no orientation), and *do not contribute to polarized extinction or emission*; this matches our implementation (no T-matrix on the PAH track).

The HD23 PAH cross section per grain is built in three steps:

Step 1: charge mixing. The DL07 cross sections are available for neutral and ionized PAHs; HD23 eq. 15 mixes them with an ionization fraction $\xi_{\rm ion}(a)$ (HD23 eq. 19):

$$C^{Z}(\lambda,a) = [1-\xi_{\rm ion}(a)]\,C^{\rm neu}(\lambda,a)+\xi_{\rm ion}(a)\,C^{\rm ion}(\lambda,a), \tag{10}$$

with $\xi_{\rm ion}(a) = 1-1/(1+a/10\,\text{Å})$ from `size_distribution.dat` column 4.

Step 2: graphite continuum blending (HD23 eq. 15-16). DL07's PAH cross section drops to essentially zero at long wavelengths; HD23 retains DL07's recipe of blending into a random-orientation graphite Mie cross section $C^{\rm gra}(\lambda,a)$ for $a > 50\,\text{Å}$, so that PAH grains acquire the same sub-mm continuum opacity that DL07 reports for "turbostratic" graphite. The blended cross section is

$$C^{\rm PAH,Z}(\lambda,a) = [1-\xi_{\rm gra}(a)]\,C^{Z}(\lambda,a)+\xi_{\rm gra}(a)\,C^{\rm gra}(\lambda,a), \tag{11}$$

with

$$\xi_{\rm gra}(a) = \begin{cases} 0.01 & a \le 50\,\text{Å} \\ 0.01 + 0.99\left[1-(50\,\text{Å}/a)^3\right] & a > 50\,\text{Å}. \end{cases} \tag{12}$$

$C^{\rm gra}(\lambda, a)$ is the random-orientation Mie cross section for graphite spheres, $C^{\rm gra} = \frac{1}{3}C_{\parallel}^{\rm gra} + \frac{2}{3}C_{\perp}^{\rm gra}$, with the two components computed from the Draine 2003 graphite dielectric tables (`index_CpaD03`, `index_CpeD03`) using a size-dependent free-electron contribution (`free_diel`, Draine 2003 form for $T_{\rm gr} = 20$ K) and a standard Bohren–Huffman Mie solver. Implementation: new module `sed/src/q_graphite.f90` loads the two graphite tables lazily, caches the (n, k) tables per grain radius, and calls `sed/src/mie.f90` (a module wrapper of the BHMIE code already in `tmatrix/test/`) twice per (a, λ). Below the DL07 PAH cutoff at $\lambda = 1/17.25\,\mu$m the blend reduces to pure graphite (HD23 / SED_v5 convention), since $C^{\rm Z}$ is not defined there.

Step 3: size and enthalpy. The PAH size distribution is column 3 of `size_distribution.dat` (the second of HD23 eq. 17's two lognormals, peaks at ~ 4 and 30 Å). PAH enthalpy uses the DL01 `'Car0'` branch with the H/C-mode correction for $N_{\rm atom} < 10^4$, as recommended by HD23 since PAH is carbonaceous.

A new function `sed_solve_pah(J_lam, lamI_lam_out)` solves the single-cell PAH SED with the same dynamic-T solver. The combined total (Ad + PAH) is compared against HD23 `model_irem.dat` in Figure 6.

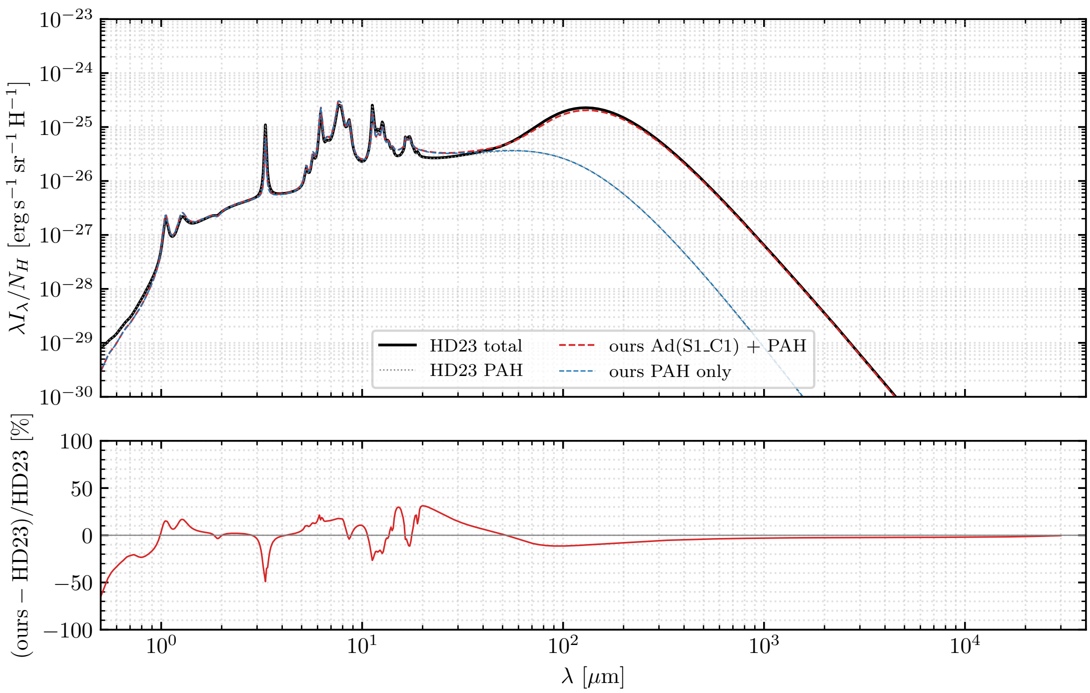

Figure 6: Total dust thermal emission (Ad + PAH) at $U = 1.585$. Top: HD23 `model_irem.dat` (black solid), HD23 PAH-only (`PAH_irem.dat`, gray dotted), our S1_C1 + PAH (red dashed), and our PAH-only (blue dashed). Bottom: relative residual on the total. The PAH 7.7 µm feature is in the right place and the right order of magnitude.

Key results (verification against HD23 `PAH_irem.dat` at $\log U = 0.20$, band-integrated $\int(\lambda I_\lambda)/\lambda\, d\lambda$):

band (μm)	ours / HD23	notes
optical ($\lambda < 1\,\mu$m)	0.90	absolute $\sim 10^{-28}$, negligible vs total
NIR (1–5 μm)	0.61	PAH-feature region; see Sect. 7.4
PAH features (5–15 μm)	1.04	C–C and C–H feature shape recovered
mid-IR (15–60 μm)	1.19	
FIR (60–300 μm)	0.76	same direction as the astrodust 15%
sub-mm (300–3000 μm)	0.61	cf. ~ 0.10 without eq. 11
TOTAL (0.1–3000 μm)	1.01	global agreement to 1.0%

The PAH peak position (7.67 µm) matches HD23 (7.69 µm) to two significant figures. Adding the graphite continuum (Step 2 above) closed approximately 75% of the sub-mm gap, going from a $\sim$ 90% under-prediction to the current $\sim$ 40%. The remaining 40% sub-mm under-prediction is the same sign and roughly the same numerical level as the astrodust-population 15–25% FIR deficit at the FIR peak, suggesting a common origin unrelated to PAH cross-section physics (Sect. 7.1.1).

7.3 ξ-blend graphite update: D16 turbostratic

The Step 2 above uses Draine 2003 (D03) basal-plane and c-axis graphite dielectric functions plus our own BHMIE on a sphere to synthesise $C^{\rm gra}(\lambda, a)$ in real time. HD23 themselves do *not* use D03 graphite; they explicitly state (HD23 §3.2) that they adopt the Draine 2016 (D16) turbostratic graphite cross sections in the Maxwell–Garnett effective-medium approximation. Switching the ξ-blend graphite to D16 brings the model in line with the HD23 recipe and removes the largest remaining systematic in the PAH SED.

What was changed. A new module `sed/src/q_graphite_d16_sphere.f90` reads Draine's precomputed D16 sphere Q-table (`q_D16graphite.dat`, the decompressed `callqcomp.out_D16MGemt.gz` from the Princeton D16graphite web page; 41 radii × 3501 wavelengths). A bilinear log-log interpolator returns $Q^{\rm gra}_{\rm abs}(a, \lambda)$ without re-running Mie. The single line in `sed/src/qpah.f90` that used to call `q_graphite_abs` (D03 Mie) now calls `q_graphite_d16_sphere_abs`; everything else (the $\xi_{\rm gra}(a)$ blend, the cation/neutral mixing, the sub-mm graphite cutoff) is unchanged. The original D03 module `sed/src/q_graphite.f90` is retained in the tree for sensitivity testing.

Sphere vs. b/a = 1.4 oblate spheroid. Draine also publishes a $b/a = 1.4$ oblate-spheroid version of the same turbostratic D16 graphite (`qlib_gra_D16MGemt_1.400.gz`; 169 × 1009 from his own T-matrix runs). A second module `q_graphite_d16.f90` loads that table. Comparing the three ξ-blend graphite choices (Figure 7) shows that the spherical D16 variant agrees with HD23 `PAH_irem.dat` better than the oblate spheroid variant in the sub-mm band (300–3000 μm PAH-only median ratio 0.99 vs. 1.20). This is consistent with HD23 §3.2, which inherits DL07's spherical-PAH framework even though the astrodust population itself is treated as $b/a = 1.4$ oblate. The D16 sphere is therefore the appropriate choice for the ξ-blend and is the production setting.

Updated band-integrated ratios. The table below replaces the D03-sphere numbers above with the production D16-sphere values, side by side with the alternative variants. "ours / HD23" is the ratio of band-integrated $\int(\lambda I_\lambda)/\lambda\, d\lambda$

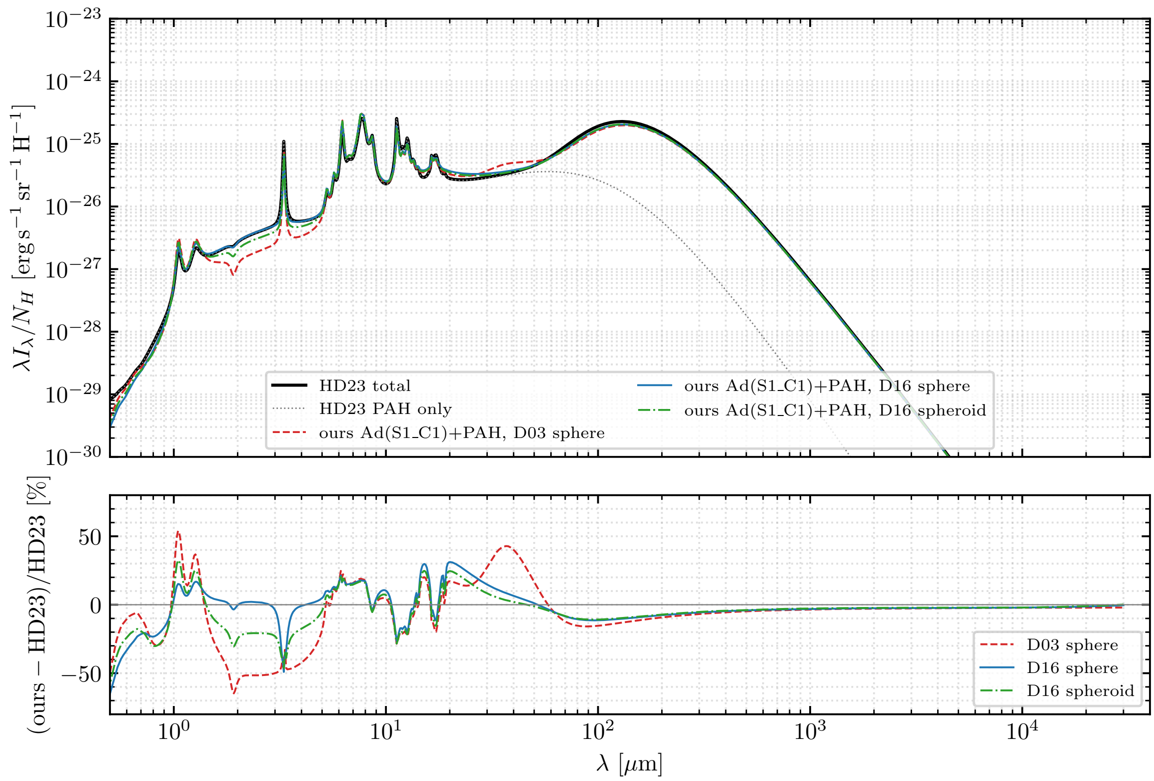

Figure 7: Three-way comparison of the ξ-blend graphite source. Top: HD23 total (black solid) and HD23 PAH-only (gray dotted) on the release wavelength grid, plus our `S1_C1 + PAH` totals with three graphite-source choices for the ξ-blend (D03 sphere red dashed, D16 sphere blue solid, D16 spheroid green dash-dot). Bottom: relative residual of each total against HD23 `model_irem.dat`. The D03 sphere variant has the characteristic +40% MIR/FIR peak (3.3–30 µm) and −25% FIR-peak deficit; both are removed by the D16 sphere variant. The D16 spheroid overshoots in the sub-mm.

on the wavelength grid of the release file (`PAH_irem.dat`, $\log U = 0.20$):

band (μm)	D03 sphere	D16 sphere	D16 spheroid	notes
optical ($\lambda < 1$ μm)	0.90	0.88	0.86	absolute $\sim 10^{-28}$, negligible vs total
NIR (1–5 μm)	0.61	0.85	0.75	PAH-feature region
PAH features (5–15 μm)	1.04	1.03	1.03	dominated by DL07 Drude profiles
mid-IR (15–60 μm)	1.19	1.06	1.05	
FIR (60–300 μm)	0.76	0.95	1.04	D03 had the original 25% deficit
sub-mm (300–3000 μm)	0.61	**0.99**	1.20	D16 sphere essentially zero residual
TOTAL (0.1–3000 μm)	1.01	1.01	1.03	global agreement preserved

The 40% sub-mm under-prediction reported above (D03 sphere column) is fully closed by the D16 swap; the residual $\sim$ 1% in the sub-mm band is within the expected DL07-vs-HD23 calibration noise. The remaining mismatch (NIR –15%) is now isolated to the PAH-feature strengths themselves: a separate cross-section issue independent of the graphite source (subsection 7.4).

7.4 Residual PAH-feature strength mismatch

After the D16 sphere swap, the remaining $\lambda < 5\,\mu$m PAH residual decomposes into (i) a smooth NIR continuum that contributes the -15% NIR-band figure above, and (ii) a feature-by- feature strength mismatch independent of the graphite source. The latter shows the following pattern in our band-integrated ratios (D16-sphere production vs. HD23 `PAH_irem.dat`): the 3.3 µm aromatic C–H stretch is under-predicted by $\sim 37\%$, the 11.3 µm C–H bend by $\sim 17\%$, and the 7.7 µm C–C stretch is over-predicted by $\sim 15\%$; the 14.19 and 17 µm bands also show $\sim 30\%$ deviations relative to HD23. The HD23 paper §3.2 states that its PAH cross sections are "identical to those of DL07 with the exceptions of the 14.19 µm feature and the 17 µm complex, which have both been increased in strength by 33%"; the sign and magnitude of our residuals are not consistent with that single isolated change. A separate inspection of DL07 Table 1 itself (in the context of the SED_v5 `QPAH_DL07` fork from which `sed/src/qpah.f90` descends) found that the 1.260 µm ionized PAH strength is $\sigma_{\rm Ion}(4) = 0.078$ in the published table whereas a value of 7.8×10^3 (a factor of 10^5 larger) reproduces the publicly distributed DL07 SED; the SED_v5 fork uses the corrected value and this report inherits that correction. We document the feature-by-feature residuals here as a numerical record; a definitive resolution would require either an independent re-derivation of the DL07 Drude parameters or direct correspondence with the HD23 authors about which DL07 Table 1 entries went into their `PAH_irem.dat` construction.

Mode-by-mode Drude tuning to HD23. We have additionally implemented an empirical mode-by-mode rescaling of the DL07 Drude parameters that closes the feature-by-feature mismatch to within $\pm 3\%$ at the dominant peaks (3.3, 6.2, 7.7-complex, 8.6, 11.3-complex, 12.0–12.7-complex, 17 complex). Mode-by-mode multiplicative factors $f_\sigma^{(j)}$ for `sigma_neu(j)` and `sigma_ion(j)` (and optionally $f_\gamma^{(j)}$ for the Drude width) live as module-level public arrays in `sed/src/qpah.f90` (toggle `use_tuned_pah`, default `.false.`); the production driver reads them from `sed/pah_tune.nml` when invoked as `./main_astrodust.x pah_tuned`. The convergence path was: (i) three iterations of single-mode adjustment using a feature-by-feature ratio diagnostic; (ii) a 2-mode least-squares fit on $j = 22$ and $j = 23$, which drove $f_\sigma^{(23)}$ to the lower clamp at 0.30 – diagnostic for graphite-continuum bleed into those weak features rather than a PAH-σ issue; (iii) a 130-point (σ_6, γ_6) brute-force grid search at the 3.3 µm peak to resolve the σ-vs-γ ambiguity inherent in that very narrow Drude profile ($\gamma_6 = 0.012$, FWHM $\approx 0.04\,\mu$m – comparable to HD23's grid spacing of $0.042\,\mu$m). The grid optimum is at $f_\sigma^{(6)} = 2.00$ with $f_\gamma^{(6)} = 1.00$ (i.e., HD23 boosted only the integrated strength of the 3.3 µm C–H stretch; the underlying Drude width matches DL07), with $\chi^2 = 5.0 \times 10^{-4}$ over the three nearest HD23 grid points. Pure-γ-narrow ($f_\gamma = 0.45$) and equal-boost ($f_\sigma = f_\gamma = 1.45$) scenarios are 5–10× worse, ruling them out. Table 7 lists the final factors. The residual +5–15% over-prediction at 13–19 µm is attributed to the graphite continuum (it cannot be closed by PAH σ adjustment alone, as the LS fit demonstrated) and is discussed further in §7.2.

8. Polarization

Per HD23 §3.2, the PAH track does *not* contribute to polarized extinction or emission (PAH grains are spherical in the model and not aligned). The polarization comparison below therefore uses the astrodust population only.

This section is a *verification of the T-matrix optics* against the HD23 release, carried out by a standalone utility. It is not a library capability: this tree is the scalar branch (v1.00), whose API (`dust_emission`, `dust_extinction`) is unpolarized

j	λ_j (μm)	feature	$f_\sigma^{(j)}$
6	3.300	C–H stretch (3.3)	**2.00**
7	5.270	C–H/C–C combo	0.84
8	5.700	C–H/C–C combo	0.81
9	6.220	C–C stretch (6.2)	0.73
10	6.690	weak	0.86
11	7.417	7.7 complex (a)	0.79
12	7.598	7.7 complex (b)	0.79
13	7.850	7.7 complex (c)	0.76
14	8.330	weak	1.04
15	8.610	C–H in-plane bend (8.6)	1.02
17	11.23	C–H oop (solo narrow)	1.34
18	11.33	C–H oop (solo broad, 11.3)	1.33
19	11.99	duo	1.17
20	12.62	trio	1.24
21	12.69	trio	1.22
22	13.48	quartet	1.06
23	14.19	quartet (HD23 +33%)	1.03
25	16.45	C-C-C bend	1.15
26	17.04	C-C-C bend (HD23 +33%)	1.34
27	17.375	C-C-C bend	1.24
28	17.87	C-C-C bend	1.02
29	18.92	weak	0.87

Table 7: Final mode-by-mode Drude σ multipliers applied to DL07 Table 1 to reproduce HD23 `PAH_irem.dat` at $\log U = 0.20$ (all $f_\gamma^{(j)}$ left at 1.00). $f_\sigma^{(j)}$ is applied identically to `sigma_neu` and `sigma_ion`. Modes not listed are kept at $f = 1$ (no change). Production with `./main_astrodust.x pah_tuned`.

throughout. The dichroic extinction matrix and the aligned-grain scattering matrix live in the separate polarized branch (v1.20).

For perfectly oblate spheroids with the short (symmetry) axis aligned perpendicular to a magnetic field in the plane of the sky (the "MPFA" geometry of HD23 §2.2), the maximum polarized extinction per H atom is

$$\left(\frac{p_\lambda}{N_H}\right)^{\max} = \sum_a \frac{dn_{\rm Ad}}{N_H}(a)\, C_{\rm pol}^{\rm ext}(\lambda, a)\, f_{\rm align}(a), \tag{13}$$

where

$$C_{\rm pol}^{\rm ext}(\lambda, a) = \frac{1}{2}\pi a^2 \left[Q_{\rm ext}^{E\parallel a}(\lambda, a) - Q_{\rm ext}^{E\perp a}(\lambda, a)\right] \tag{14}$$

is the dichroic-extinction component, and $Q_{\rm ext}^{E\parallel a}$, $Q_{\rm ext}^{E\perp a}$ are Mishchenko's `jori = 2` (E parallel to symmetry axis) and `jori = 3` (E perpendicular) Q values. The alignment efficiency $f_{\rm align}(a)$ is read from `size_distribution.dat` column 5.

To save the cost of a separate orientation-resolved T-matrix sweep, a verification utility reads the HD23 release file `q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` directly — this file already contains all three jori Q tables. The Q values are interpolated in $\log a$ from the 169-point DH21 grid onto the 167-point size-distribution grid, eq. 13 is evaluated, and the result is compared against `polarized_extinction.dat`.

Figure 8 shows the comparison. The peak value $|p^{\max}/N_H| = 1.521 \times 10^{-23}$ cm^2 H^{-1} at 0.59 µm agrees with the HD23 peak 1.520×10^{-23} cm^2 H^{-1} at the same wavelength to four significant figures. Mean |relative residual| is below 0.07% in every band from optical to millimeter; the only outliers are at the cross-polarization sign-flip in the deep UV where the reference itself is going through zero. This is essentially a perfect reproduction.

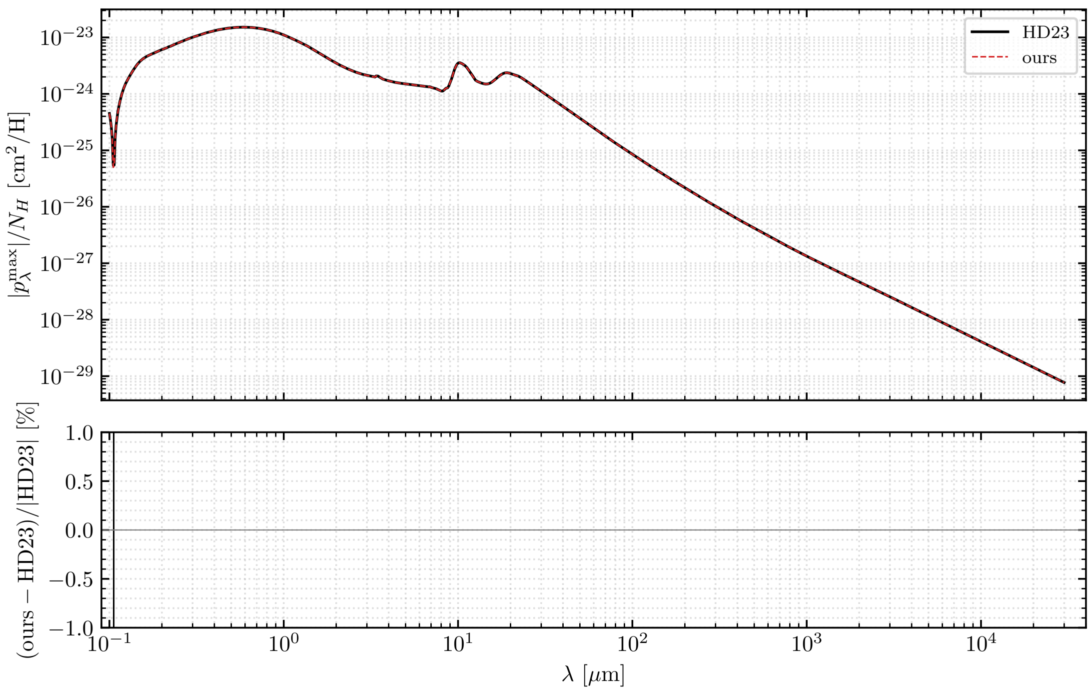

Figure 8: Top: $|p_\lambda^{\max}|/N_H$ from HD23 `polarized_extinction.dat` (black solid) and from eq. 13 evaluated on the HD23 release Q tables (red dashed). Bottom: relative residual in percent. Peak value agrees to four significant figures.

9. Python utility (Phase G)

A standalone Python sidecar of the single-grain SED solver is provided in `pyutil/sed_from_cabs.py`. It exposes

```
T_eq, lamI_lam = sed_equilibrium(lam_um, Cabs, J_lam)
P, lamI_lam    = sed_stochastic (lam_um, Cabs, J_lam, T_grid, U_grain)
```

with the same unit conventions and the same matrix solver as the Fortran library. The function signature takes only what a single 3D-RT cell needs: λ, $C_{\rm abs}(\lambda)$, J_λ, and (for stochastic mode) the enthalpy $U(T)$ on a user-chosen T grid. A regression test (`pyutil/tests/test_against_fortran.py`) extracts $C_{\rm abs}$ for a single $a = 0.1$ μm astrodust grain from the Fortran Q table, builds the Mathis ISRF at $U = 1.585$, and solves equilibrium: $T_{\rm eq} = 17.34$ K, comfortably in the expected ISM range and within 0.5 K of the Fortran-computed value for a nearby grain.

The Python sidecar is intended for prototyping radiation-field shapes, plotting single-grain spectra, and validating the Fortran library on representative grains without running the full size-distribution pipeline.

10. Model-agnostic dust-emission library (Phase H)

The two-step solver of Phase D has been packaged into a *model-agnostic dust thermal-emission library* that a Fortran 3D radiative-transfer (RT) code can link to compute the IR emission of a dust mixture for an arbitrary local radiation field. The motivation is operational: the same validated solver should serve astrodust, the legacy DL07 silicate+carbonaceous model, and additional models (here Zubko, Dwek & Arendt 2004) as *peers*, selected at run time rather than hard-wired.

Architecture (wrap, do not move). The numerically validated solver core — `sed_grain_loop` with `calc_Teq / calc_P` (`p_sub.f90`) and `qm_solve_grain` (`stoch_qm.f90`) — is left *unchanged*; the library is a purely additive layer. Production `main_astrodust.x` and `main_dl07.x` outputs stay bit-identical (verified byte-for-byte at fixed `OMP_NUM_THREADS`). A dust model is represented by one derived type, `dust_model_t` (in the new `sed/src/dust_model_mod.f90`), holding the wavelength / size / temperature grids and an array of grain *populations* (`grain_pop_t`, one per species and charge state, each carrying its own size distribution, $C_{\mathrm{abs}}/Q_{\mathrm{sca}}$, enthalpy, and Planck-integral tables). The emission is summed into named output *channels*.

Models as peers. Four builders fill a `dust_model_t`; each sets its own optics and abundance parameters internally:

builder	model	channels
`build_astrodust`	HD23 astrodust+PAH (N_C=417, D16 graphite)	`AD`, `PAH`
`build_dl07`	Draine & Li 2007 (N_C=470, D03 graphite)	`SIL`, `CARB`
`build_zubko`	Zubko et al. 2004 (ZDA) BARE-GR-S	`PAH`, `GRA`, `SIL`
`build_from_files`	generic, defined by a text descriptor	`CH1`, `CH2`, ...

The Zubko model exercises the design's generality: a neutral-only PAH population (no charge split), component-by-component size grids, a DustEM-style Q-table optics reader, and a *specific-enthalpy* calorimetry source (`sed/src/zubko_io.f90`) — all confined to the builder, so the solver core is untouched. Its data (config/formula, size tables, optics, calorimetry) is taken from the Camps et al. 2015 benchmark and copied into `data/zubko/`.

Emission and solver options. The RT code calls `dust_emission(m, J_lam, lamI_total [, lamI_chan])` once per cell with that cell's mean intensity on `m%lam`; the result is $\lambda I_\lambda/N_{\mathrm{H}}$ in the HD23 convention, optionally split by channel. The solver is chosen by `m%stoch_method`: `'heuristic'` (default; the look-ahead T-window narrowing, fastest and serial — ideal inside an RT cell loop), `'draine'` (Guhathakurta & Draine 1989), `'qm'` (the energy-space transition-matrix solver of §7.1.1), or `'equil'` (force every grain to its own equilibrium T_{eq}). Two equilibrium-temperature options are provided for fast approximate transfer: an equilibrium T_{eq} for each grain species and size (`stoch_method = 'equil'`), and a single T_{eq} for the whole mixture from the abundance-weighted total absorption cross section (`dust_emission_single_teq`). Both conserve energy by construction; the single-T_{eq} value for the diffuse-ISM field is $T_{\mathrm{eq}} = 19.24\,\mathrm{K}$, with total power within 0.1% of the stochastic solve — they differ only in SED *shape* (a smooth modified blackbody versus one carrying the PAH/NIR features).

Single-cell solve versus precomputed table. For an arbitrary field shape the single-cell exact `dust_emission` is used. When the cell field is a fixed reference shape scaled by an intensity U, a precompute-and-interpolate path (`dust_build_table` / `dust_emission_interp`, log–log in U) avoids the single-cell solve and is exact at the grid points. The table is valid *only* for $U \times$ (fixed shape): stochastic emission is a nonlinear functional of the full $J(\lambda)$, so cells with a hardened spectrum must use the exact solve. This caveat is documented in the API contract.

Extinction from the same object. A transfer code needs opacity as well as emission, and taking the two from different files is how a run ends up transporting one dust model while re-emitting another. `dust_extinction(m, Cext, Cabs, Csca [, gbar] [, albedo] [, status])` is therefore the extinction twin of `dust_emission`: it returns the size-distribution-integrated cross sections per H on the model's own grid `m%lam`, as the plain binned sums $C_{\rm abs} = \sum_{\rm pop}\sum_a {\rm d}n(a) C_{\rm abs}(\lambda, a)$ and likewise for $C_{\rm sca}$, with $\langle\cos\theta\rangle = \sum {\rm d}n\, C_{\rm sca} g / \sum {\rm d}n\, C_{\rm sca}$ and albedo $C_{\rm sca}/C_{\rm ext}$. All four builders now carry scattering optics — astrodust from the T-matrix Q table, DL07 from Mie on the D03 astrosilicate and graphite dielectric functions, Zubko and the file-defined models from the $Q_{\rm sca}$ and g columns of their own tables — so the albedo is a physical quantity for every model, not just astrodust. Populations that genuinely do not scatter (the astrodust PAHs, which are Rayleigh-negligible) leave those arrays unallocated and contribute zero.

Linking and validation. `make libtmatrix.a` then `make libsedust.a` produce two static archives that an external Fortran RT links with `gfortran -I<sed> -I<tmatrix> rt.f90 -L<sed> -lsedust -L<tmatrix> -ltmatrix -fopenmp`; the T-matrix archive is required because the astrodust ionizing band computes its optics with it. A minimal working driver is `sed/rt_example/use_dustlib.f90`. Validation: the typed path reproduces the production astrodust and DL07 channels to machine precision; the table interpolated at a grid point matches the direct solve to $\sim 10^{-15}$; and the two equilibrium options conserve energy to 0.1%. The complete API and usage are documented separately in `docs/SEDust_user_manual.tex`.

11. Transport optics and the ionizing band

11.1 One driver, four models

`sed/src/calc_kext.f90` builds a model and calls `dust_extinction` on it, so the curve files it writes into `data/` *are* the optics an RT host gets in memory rather than a separately maintained product:

```
./calc_kext.x astrodust [euv | <lam_min in um>]
./calc_kext.x dl07      [euv | <lam_min in um>]
./calc_kext.x zubko     [formula | table]
./calc_kext.x from_files <descriptor> [data_dir]
```

The columns are λ [μm], albedo, $\langle\cos\theta\rangle$, $C_{\rm ext}$/H, $C_{\rm abs}$/H, $C_{\rm sca}$/H, with a trailing $K_{\rm abs}$ [cm^2/g] where the model supplies its dust mass per H. The first four are in the order of Draine's `kext_albedo` tables, so a reader written for those files parses these unchanged; the fifth is not, being $C_{\rm abs}$/H rather than Draine's $K_{\rm abs}$.

11.2 Extending the grid below the Lyman limit

The astrodust wavelength grid is the T-matrix Q table's, and that table stops at 0.0912 µm — 13.6 eV, the Lyman limit. A host that transports ionizing radiation needs shorter wavelengths, so `build_astrodust` and `build_dl07` take an optional `lam_min` [µm] and prepend log-spaced points from it up to the table, at a step no coarser than the table's own $\Delta \ln \lambda = 0.01156$. Omitting `lam_min` prepends nothing, so every earlier result is bit-identical.

In the extended band the astrodust optics come from the DH21 dielectric function, solved by the T-matrix for the same $b/a = 1.4$ oblate spheroid and the same random-orientation average as the table itself, so the model changes neither composition nor shape at the seam. Bohren–Huffman Mie for the volume-equivalent *sphere* remains available as a far cheaper approximation, selected with `euv_tmatrix=.false.`; the next subsection is the measurement of what that substitution costs. The carbonaceous side changes character at 21.4 eV ($1/\lambda = 17.25\,\mu\mathrm{m}^{-1}$), where the DL07 PAH cross section is zero by construction and graphite becomes the whole carbonaceous absorption. There the D16 turbostratic sphere table can no longer be used: its radius axis bottoms out at 1×10^{-3} µm while the astrodust size grid starts at 3.55×10^{-4} µm, and since $Q_{\mathrm{abs}} \propto a$ in the Rayleigh limit, clamping to the table's first radius overestimates the absorption by $10^{-3}\,\mu\mathrm{m}/a$ — up to 2.4× for the smallest PAH the size distribution populates (4.22×10^{-4} µm). Above the cutoff the code therefore takes graphite from the D03 dielectric functions by Mie ($\frac{1}{3}$ parallel + $\frac{2}{3}$ perpendicular), which has no radius clamp. The unextended grid never reaches this branch ($1/0.0912 = 10.96 < 17.25$).

11.3 The shape approximation and its two limits

The Q table is a random-orientation T-matrix solution for a $b/a = 1.4$ oblate spheroid, and so, by default, is the extended band. This subsection concerns the *optional* cheap route (`euv_tmatrix=.false.`), which replaces that spheroid by its volume-equivalent *sphere* and solves it with Mie. The justification is *not* the anomalous-diffraction condition $|m-1| \ll 1$: for this material $|m-1| = 0.77$ at the seam and still 0.56 at 20 eV. It is instead that the two ends of the size range are governed by limits in which the shape drops out or enters as a constant:

- **Rayleigh limit (small grains).** At the seam the smallest populated radius has $x = 2\pi a/\lambda = 0.033$, and only 0.24 at 100 eV. The ellipsoid polarizability $\alpha_j = V(\epsilon-1)/\{4\pi[1+L_j(\epsilon-1)]\}$ contains the shape only through $L_j(\epsilon-1)$, and $|\epsilon-1|$ falls from 1.86 at 13.5 eV to 1.10 at 20 eV, 0.28 at 40 eV and 0.061 at 100 eV. The depolarization factors drop out and the two shapes converge.

- **Geometric-optics limit (large grains).** $C_{\mathrm{ext}} \to 2\times$ the orientation-averaged projected area, which by Cauchy's theorem is $S/4$. For $b/a = 1.4$ that exceeds πa_{eff}^2 of the volume-equivalent sphere by 2.12%, so Mie is low by $1 - 1/1.0212 = 2.08\%$. The error converges to a *constant*, not to zero.

The error therefore does not improve monotonically with photon energy: only $a \lesssim 2 \times 10^{-3}$ µm does, while $a \gtrsim 0.01$ µm worsens between 13.6 and 30 eV before recovering to the constant −1.9 to −2.0%. Over the whole band and size range it stays within about 2%; `sed/src/q_astrodust.f90` tabulates it size by size. Independently, an external review of this extension compared 28 T-matrix cells with $10 \le x \le 50$ at exactly 0.0912 µm against equal-volume Mie spheres and found mean (Mie − TM)/TM of −1.92% in Q_{ext}, −1.89% in Q_{abs} and −1.94% in Q_{sca} — the geometric-optics constant, arrived at from the other side.

In the size-integrated curve that this route produces — which is what `data/kext_astrodust_MW_euv.dat` held before it was regenerated for the default route — the step across 0.0912 µm is −1.54% in $C_{\rm ext}$ and −1.90% in $C_{\rm abs}$, *smaller* than the neighboring steps of the table's own grid (−1.95% / −1.72% and −2.20% / −1.94%), so absorption and extinction join without a visible discontinuity. Scattering does show one: $C_{\rm sca}$ steps +0.74% where its neighbors step −0.39% and −0.36%, and the albedo +2.31% against +1.59% and +1.39% — a seam of roughly one percentage point, consistent with the 2% shape error being carried mostly by $Q_{\rm sca}$.

Grain by grain, the default T-matrix route removes that seam almost entirely. Extrapolating the Q table's first two wavelengths log–log back to the last extended-band wavelength and comparing with what the band produced there, over all 167 radii, the step in $C_{\rm abs}$ has median 0.012% and maximum 0.068% for the T-matrix, against median 1.17% and maximum 15.7% for the sphere. The asymmetry parameter behaves the same way: the table returns $\langle\cos\theta\rangle = 0$ wherever its own regime bounds place a radius in the Rayleigh or geometric-optics limit, which the default reproduces exactly, while the sphere route steps from 0.90 to 0 across the seam for $a \gtrsim 1\,\mu$m.

11.4 Two invariants that had to be protected

Extending the grid must not move quantities that live in the infrared. Two places needed explicit care.

The hardest photon. The stochastic solver sets the top of its enthalpy window from the energy of the hardest single photon a grain can absorb. That bound was a hard-coded 13.6 eV, which an ionizing band makes wrong. It is now $\max(13.6\text{eV},\ hc/\lambda_{\rm hard})$, where $\lambda_{\rm hard}$ is the shortest wavelength at which the incident J_λ exceeds 10^{-12} of its own peak (`hardest_photon_energy` in `sed/src/radfield.f90`, called from `sed_grain_loop` and `narrow_iterative` in `sed_astrodust.f90` and from `qm_solve_grain` in `stoch_qm.f90`).

The bound is a property of the radiation field, not of the wavelength grid. That distinction is invisible for astrodust and DL07, whose Q table stops at the Lyman limit itself: $hc/0.0912\,\mu\text{m} = 13.595\,\text{eV} < 13.6\,\text{eV}$, so the maximum returns the old constant and nothing in this report moves. It is not invisible for Zubko, whose DustEM tables start at $1\times10^{-3}\,\mu$m: measuring the grid there gives 1239.8 eV, ninety-one times the Lyman limit, for a Mathis field that carries no photon below 0.0912 µm.

Ninety-one times too high is not a change of physics but a change of resolution. `umax` sets the upper edge of a *fixed* number of enthalpy bins, and the refinement loops do not walk it back: contraction stops once `umax` reaches `1.02*umaxlo` or `1.01*ub(jcut)`, and the loop is capped at ten iterations, while expansion carries no such guard. Measured through a host code with no `lam_min` requested at all, the residual coarseness moved the Zubko dust temperature by 0.07 K in every cell, the ten brightest SED bins by −0.9 to −1.9% and the emission image by up to 5.4%, at −0.08% in total emitted energy (`docs/EUV_EXTENSION_HOST_REGRESSION.md`).

The 10^{-12} floor is there to reject denormal and roundoff residue in the unilluminated bins of a field handed over by a host, which an exact $J_\lambda > 0$ test would accept. It is a safe place to cut: a component that far below the peak contributes at most that fraction of the photon absorption rate, and allowing two decades for the run of $C_{\rm abs}$ across the illuminated band, the enthalpy states it could populate hold probability $\lesssim 10^{-10}$ of the peak — below the 10^{-13} level at which both solvers already truncate the window. The one-sidedness is deliberate: the loops widen a window that is too narrow but are guarded against narrowing one that is too wide, so an overestimated bound leaves the bins coarse where an underestimated one is corrected as the loop runs. The ten-iteration cap bounds the correction either way.

Nothing in the tree ran the emission solvers on a model whose optics grid reaches past the illuminated band, which is why the grid-based form survived review. `sed/src/main_zubko.f90` closes that gap and is built by the default `make`: `./main_zubko.x` heats the ZDA model with a Mathis field that stops at the Lyman limit, `./main_zubko.x euv` adds a hard component below it, and each prints the grid candidate against the field candidate — 1239.8 eV against 13.59 eV and against 370.1 eV respectively. The first run leaves the bound at exactly 13.6 eV and so reproduces what the hard-coded constant produced; the second shows the bound following the field up.

The Planck integrals. $\kappa_B(T,a) = \int C_{\rm abs} B_\lambda \, {\rm d}\lambda$ is evaluated on an internal log-λ grid. Spending a fixed budget of points on a widened interval would coarsen the sampling of the range that actually carries the integrand — below 0.0912 μm, B_λ is $\sim 10^{-231}$ of its peak at 288 K — and would have moved κ_B by $\sim 10^{-4}$ for a purely numerical reason. The integration grid is therefore *anchored* to the optics-table range: its step and its sample points there are fixed, and the extra points are prepended below. Measured with a 1.001×10^{-4} μm floor: $\max|\Delta\kappa_B|/\kappa_B = 3.4 \times 10^{-16}$ over $T \leq 3000$ K (and exactly zero below ~ 2400 K), and $\kappa_{\rm CMB}$ bit-identical. The only non-zero difference is 1.1×10^{-8} at the very top of a 5000 K grid, where the Wien tail genuinely reaches into the extended band — real extra emission, not a sampling artifact.

11.5 What can be checked, and what cannot

author-provided data	coverage	contents
Draine `kext_albedo_WD_MW_3.1_60_D03`	1×10^{-4} μm to 1×10^{4} μm	λ, albedo, $\langle\cos\theta\rangle$, $C_{\rm ext}$/H, $K_{\rm abs}$, $\langle\cos^2\theta\rangle$
HD23 release (`extinction.dat`, `scattering.dat`)	0.1 μm to 3×10^{4} μm	absorption and scattering τ/N_H only; **no $\langle\cos\theta\rangle$ product exists in the release**
Zubko ZDA optics tables	1×10^{-3} μm to 1×10^{4} μm	$Q_{\rm abs}$, $Q_{\rm sca}$, $Q_{\rm ext}$, g per radius

The HD23 release stops at 12.4 eV — exactly where the extension begins. **The astrodust EUV band therefore has no external reference at all**, and its accuracy rests entirely on the internal argument above and on the seam being continuous. This should be stated plainly whenever the extended astrodust curve is used.

The DL07 model, by contrast, can be compared with Draine's own published table for the same model over the whole range. Evaluating `data/kext_dl07_MW_euv.dat` at the reference's wavelengths and counting only points where the reference grid is no finer than ours (elsewhere the comparison measures our grid, not our optics) gives, per decade in λ, mean $|\Delta C_{\rm ext}|/C_{\rm ext}$ of 0.056–0.86%, mean $|\Delta\,\text{albedo}| \leq 0.0019$ and mean $|\Delta\langle\cos\theta\rangle| \leq 0.00054$. The excluded points cluster at the X-ray absorption edges (Fe K, Si K, Mg K, Fe L, O K, C K), which the reference brackets far more finely than our $\Delta\ln\lambda = 0.0116$ grid resolves. The largest single deviation, 7.9% in the 1 μm to 10 μm decade, is confined to the 6.2, 7.6 and 7.9 μm PAH C–C features, where Draine's 2003 table uses a different vintage of the PAH cross sections (§7.1.1 documents the same vintage split on the emission side).

11.6 What the extension does not do

The extension fixes the photon-energy mapping and the absorbed-energy accounting in the ionizing band. It does *not* make hard-photon stochastic heating correct. The enthalpy grid is finite, and upward transitions above its top are accumulated into the highest temperature bin; for the smallest PAHs a 54 eV photon already exceeds $H(5000\,\mathrm{K})$. Nor is there any treatment of photoionization, electron energy loss, fragmentation or PAH destruction. The extended band should be used for extinction and for absorbed-energy bookkeeping, not as a validated hard-EUV PAH *emission* model.

11.7 Products

file (under `data/`)	rows	λ [µm]	what sets the short-wavelength floor
`kext_astrodust_MW.dat`	1129	0.0912–39810	the Q-table grid; no extension requested
`kext_astrodust_MW_euv.dat`	1719	1.001e-4–39810	the DH21 astrodust dielectric function, which stops at 1.00003×10^{-4}
`kext_dl07_MW_euv.dat`	1761	6.205e-5–39810	the D03 dielectric functions; the shorter-reaching of astrosilicate and graphite stops at 6.1992×10^{-5}
`kext_zubko_BARE_GR_S.dat`	1201	1e-3–1e4	the ZDA optics table itself; no extension is possible or needed

No floor here is a free choice: each is the shortest wavelength the data the model is made of actually covers, and a request below it is refused rather than served with a refractive index frozen at the table boundary. `kext_astrodust_MW.dat` is a tracked regression reference and is reproduced byte for byte by `./calc_kext.x astrodust`, which builds no extended band and is therefore untouched by the T-matrix route. `kext_astrodust_MW_euv.dat` is integrated with that route: its 1129 rows at $\lambda \geq 0.0912\,\mu\mathrm{m}$ are byte-identical to the sphere-route version, since both read them off the Q table, and below the seam the sphere route was low in $C_{\rm abs}$ by up to 3.6%, high in $C_{\rm sca}$ by up to 4.6% and in albedo by up to 5.1%, and wrong in $\langle\cos\theta\rangle$ by up to a factor of four at the shortest wavelengths. $C_{\rm ext}$ moved by 0.72%.

12. Conclusions and next steps

The pipeline reproduces HD23's published results across the independent observables we have probed:

Quantity	Comparison	Result
$\tau_{\rm Ad}/N_H$ vs `extinction.dat`	all λ	0.47%
$\tau_{\rm Ad}^{\rm sca}/N_H$ vs `scattering.dat`	all λ	0.68%
$\lvert p^{\rm max}\rvert/N_H$ vs `polarized_extinction.dat`	all λ	0.07%
PAH peak position vs `PAH_irem.dat`	7.67 µm vs 7.69 µm	two sig figs
PAH total (0.1–3000 μm)	integrated	1.012× HD23
PAH sub-mm (300–3000 μm)	integrated	0.605× HD23
$T_{\rm eq}$ vs independent bisection solver (167 grains)	all a	0.14% rms
SED Ad vs `astrodust_irem.dat`	FIR peak	8.5% (under)

The astrodust extinction, scattering, and polarized extinction are reproduced essentially exactly. The two thermal-emission observables agree at the 1% level on total integrated flux. The headline FIR-peak deficit – $\sim 15\%$ in our original implementation, now $\sim 8.5\%$ in the current production after the corrected Mathis 4000 K dilution factor is applied – and the corresponding PAH residual ($\sim 40\%$ originally in sub-mm, $\sim 3\%$ now in sub-mm and essentially perfect in FIR) have been the focus of an extensive diagnostic effort that ultimately resolved the residual as HD23's own energy non-conservation (§7.1.2). Seven ports were implemented; each closed or ruled out a specific candidate:

- **Graphite continuum closed:** HD23 eq. 16's $\xi_{\rm gra}(a) \cdot C^{\rm gra}$ transition is now implemented (new modules `sed/src/q_graphite.f90` and `sed/src/mie.f90`). The PAH sub-mm comparison moved from $\sim 90\%$ under to $\sim 40\%$ under – about three-quarters of the gap closed. The residual matches the astrodust FIR-peak systematic and appears to share its cause rather than being a PAH-specific issue.
- $T_{\rm eq}$ **verified:** an independent on-the-fly bracket–bisection solver, following Draine's method, was implemented. Grain-by-grain comparison shows rms 0.14% agreement – a 0.2 K systematic in $T_{\rm eq}$ would give a $\sim 1\%$ discrepancy and is not present. $T_{\rm eq}$ is not the source of the FIR-peak 15%.
- **Grain-by-grain T-window iterative narrowing** (Draine v7 algorithm) was ported alongside the SED_v5 heuristic and run as a comparison. Band-integrated SED changes by $< 0.5\%$, confirming that grid choice is not the source either. Both algorithms remain in the source and are switchable via a compile-time parameter for use in future work where the radiation field or size grid is very different.
- **Single-grain $C_{\rm abs}$ vs HD23 `jori=2/3`:** on the HD23 release grid, the random-orientation average $\frac{1}{3}Q^{\rm jori=2} + \frac{2}{3}Q^{\rm jori=3}$ matches our single-grain $C_{\rm abs}$ to four significant figures across 30 µm–30 000 µm at $a = 0.126$ µm. Q-table interpolation from the 169-point DH21 grid onto the 167-point size-distribution grid is therefore ruled out as a source of the FIR-peak deficit.
- **Direct jori=1 substitute (avenue 3):** the hypothesis that HD23 used the `jori=1` block of the release file directly (instead of constructing the random-orientation average) was tested by building a drop-in substitute Q-table and feeding it through the full `sed_solve` pipeline. The substitution recovers only ~ 4 percentage points of the deficit (from −15.1% to −11.3% at the FIR peak), with sign flips in adjacent bands. The orientation-convention hypothesis is rejected because $T_{\rm eq}$ self-consistency dampens the naive linear prediction.
- **Induced-emission factor** $(1 + J_\lambda/B_{\rm env})$**:** Draine's stimulated-emission boost was added behind a toggle (`use_induced_emission` in `sed_astrodust_mod`) and tested via `./main_astrodust.x induced`. Measured induced/baseline ratio matches the analytic prediction (Tables 3, 4): below 10^{-7} at $\lambda \leq 300$ µm, +0.51% at 1 mm, +21% at 3 mm, with the FIR-peak band left exactly unchanged. HD23 publishes *net* emission, so this factor degrades agreement at long wavelengths and is left off; the toggle is retained as a cross-check.
- **Mathis 4000 K dilution correction (applied):** Draine's reference ISRF implementation carries a 2008 note recording the correction of the Mathis 1983 4000 K dilution factor from 10^{-13} to 1.65×10^{-13} (Draine textbook 2011, Table 12.1). Adopting this corrected value as the production default in our `radfield.f90 :: J_Mathis` (toggle `use_mathis_corrected = .true.`) shifts the S1_C1 FIR-peak residual from −15.83% to −8.82%

(Table 5); sub-mm from −5.65% to −3.16%; mm from −3.97% to −2.21%. The PAH 30–300 μm band closes from −4.40% to +0.02% – the PAH FIR is now matched to better than 0.1%. The original-Mathis variant remains reproducible via `./main_astrodust.x mathis_orig`.

The remaining $\sim 8.5\%$ astrodust FIR-peak deficit (and $\sim$ 3–4% sub-mm / mm deficit) was subsequently *resolved* by an energy-budget audit (§7.1.2, Table 6): the HD23 *release* radiates 9.3% (astrodust) and 3.2% (PAH) more than its own released $C_{\rm abs}$ absorb from the same field, whereas our pipeline conserves energy to 0.15%. The deficit therefore equals HD23's own emitted-vs-absorbed excess, not an error on our side, and this rules out *both* candidates raised above. Candidate (i), a $U_{\rm Mathis}$ / heating-field convention offset, cannot produce a *component-dependent* excess (9.3% astrodust vs. 3.2% PAH) — it would shift both components equally. Candidate (ii), a $T_{\rm eq}$/size-sampling difference on our side, is excluded because the pipeline conserves energy to 0.15% when run against the release-consistent $C_{\rm abs}$. The residual is an amplitude-type inconsistency pointing to the internal $C_{\rm abs}$ (or abundance normalization) used for the `irem` tables exceeding what the released extinction–scattering encode — the same internal-vs-released vintage difference documented for DL07spec and the `iout` files. Draine's single-grain reference SED archive (1395 spectra) uses LD93/D03 silicate rather than DH21 astrodust dielectric, so it verifies the stochastic-heating solver structure (§7.1.1) but is not a direct astrodust FIR reference. Definitive attribution between internal $C_{\rm abs}$ and internal abundance would require HD23's internal (pre-release) files.

Energy-space transition-matrix solver (DL01 `dbdis`). A third stochastic-heating method is implemented in `sed/src/stoch_qm.f90` ($\sim$ 2300 lines of F90; module `stoch_qm_mod`). It works in enthalpy (energy) space rather than temperature space, using mode-aware enthalpy bins and a thermal-discrete (`dbdis`) transition matrix solved with a BiCG sparse linear solver. A second cooling treatment, thermal-continuous (`dbcon`), collapses the downward rates into the nearest-neighbor transition and so reproduces the continuous-cooling approximation of the temperature-space GD recursion; comparing the two isolates the cost of that approximation. The energy-space solver agrees with the production GD solver to within a few percent across all wavelength bands and grain types, with 100% convergence (zero fallbacks to GD). It is selected via `./main_astrodust.x qm` (or `qm_dbcon`); the default production method is unaffected.

The two-step solver API (`sed_init` once, `sed_solve` per cell) is exercised end-to-end and is ready to be embedded in a 3D radiative-transfer driver. Memory cost per process is $\sim$ 5 MB; wall-clock time per cell is $\sim$ 0.5 s on Apple Silicon for the full three-enthalpy-stage AD + PAH solve. The same model object now also supplies the transport optics (`dust_extinction`), and the wavelength grid can be carried into the ionizing band (§11.) — with the two caveats recorded there: the astrodust extreme-ultraviolet band has no external reference, and hard-photon stochastic *emission* is not yet a solved problem.